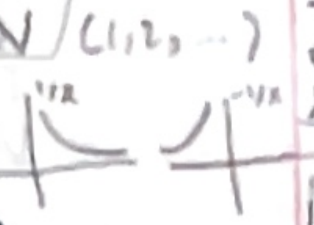
Q^c | Q — Z $(-2,-1,0,1,2)$ — N $(1,2,\dots)$



Recall

- a set can be (in)finite
- → infinite set is (un)countable
- → finite set is finite or countably infinite

} $\boxed{\mathbb{R} = Q \cup Q^c}$

Indicator Fx: $\mathbb{1}(A) = \begin{cases} 1, & A \text{ true} \\ 0, & \text{otherwise} \end{cases}$

$\boxed{A \subseteq B \Rightarrow P(A \cap B) = P(A)}$

Geometric series

$\boxed{\forall r \neq 1 \in \mathbb{R}, \sum_{k=0}^{n-1} r^k = \frac{1-r^n}{1-r}}$

$\boxed{\forall |r| < 1 \Leftrightarrow r \in (-1,1), \sum_{k=0}^{\infty} r^k = \frac{1}{1-r}}$ otherwise diverge

$\sum_{i=1}^{\infty}(1-p)^{i-1}p = p\sum_{k=0}^{x-1}(1-p)^k$, $k = i-1$ (change-of-var)

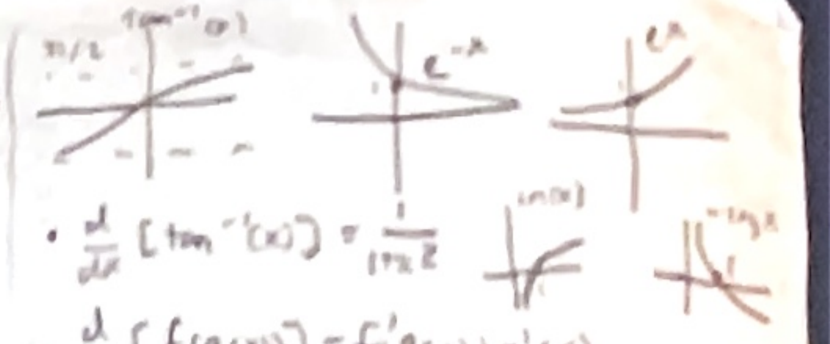
$S: B := (-\infty, a], a \in Q$

$(-\infty, b] = \bigcap_{n=1}^{\infty}(-\infty, q_n] \in \mathcal{B}$, $q_n \downarrow b$

$(-\infty, a) = \bigcup_{q<a, q\in Q}(-\infty, q] \in \mathcal{B}$

$\{a\} = (-\infty, a] \setminus (-\infty, a)$
$= (-\infty, a] \cap ((-\infty, a))^c \in \mathcal{B}$

$(a,b] = (-\infty, b] \setminus (-\infty, a)$
$= (-\infty, b] \cap ((-\infty, a))^c \in \mathcal{B}$



- $\frac{d}{dx}[\tan^{-1}(x)] = \frac{1}{1+x^2}$
- $\frac{d}{dx}[f(g(x))] = f'(g(x))g'(x)$
- $e^A e^B = e^{A+B}$
- $(e^A)^B = e^{AB}$

(★) Generalizes $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$

1.1: Set Theory

sample space (S): the set of all possible outcomes of an experiment
→ can be (un)countable; dictates prob. assignments
countable: $s \in S$ can be put into a 1-1 correspondence w/ subset of integers

Event: any collection of possible outcomes of an experiment
→ any subset of S, including S, i.e. $E \subseteq S$.

$\boxed{A \subset B \Leftrightarrow x \in A \Rightarrow x \in B}$

$\boxed{A = B \Leftrightarrow A \subset B \text{ and } B \subset A}$

Elementary set ops.
union, intersection, complementation

add A_i in practice

$\boxed{A \setminus B = \{x \mid x \in A, x \notin B\}}$ $A \cap B^c$

$\boxed{A \cup B = \{x : x \in A \text{ or } x \in B\}}$

multiply A_i in practice — $\boxed{A \cap B = \{x : x \in A \text{ and } x \in B\}}$

$\boxed{A^c = \{x : x \notin A\}}$ $\emptyset$ - empty set

Disjoint (mutually exclusive): $:= \boxed{A \cap B = \emptyset}$ Pairwise Disjoint $:= \boxed{A_i \cap A_j = \emptyset \ \forall i \neq j}$

Partition (of S): $A_1, A_2, \dots$ pairwise disjoint and $\bigcup_{i=1}^{\infty} A_i = S$.

Properties

commutativity	$A \cup B = B \cup A$ $A \cap B = B \cap A$
Associativity	$A \cup (B \cup C) = (A \cup B) \cup C$ $A \cap (B \cap C) = (A \cap B) \cap C$
Distributivity	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
Demorgan's (law)	$(A \cup B)^c = A^c \cap B^c$ $(A \cap B)^c = A^c \cup B^c$

countable union (existence: ∃) w at least one A_i

$\bigcup_{i=1}^{\infty} A_i = \{x : x \in A_i \text{ for some } i < \infty\}$
$= \{x \in S : x \in A_i \text{ for some } i\}$
$= \lim_{n\to\infty} \bigcup_{i=1}^{n} A_i$

countable intersection (∀ - for all)

$\bigcap_{i=1}^{\infty} A_i = \{x : x \in A_i \text{ for all } i = 1, 2, \dots\}$
$= \{x \in S : x \in A_i \text{ for all } i\}$
$= \lim_{n\to\infty} \bigcap_{i=1}^{n} A_i$

$\boxed{\left(\bigcup_{i=1}^{\infty} A_i\right)^c = \bigcap_{i=1}^{\infty} A_i^c}$

1.2: Probability Theory Basics | 1.2.1: Axiomatic Foundations

Def: a collection on subsets of S is a sigma algebra (Borel field), $\mathcal{B}$ if it satisfies:

i) $\boxed{\emptyset \in \mathcal{B}}$ ii) $\boxed{A \in \mathcal{B} \Rightarrow A^c \in \mathcal{B}}$ iii) $\boxed{A_1, A_2, \dots \in \mathcal{B} \Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{B}; \bigcap_{i=1}^{\infty} A_i \in \mathcal{B}}$

- $S \in \mathcal{B}$; $\{\emptyset, S\}$:= trivial sigma algebra

Q: How to show that sets are $\in \mathcal{B}$?
A: verify the set in question can be obtained from known elements of $\mathcal{B}$ via countably many set operations
note: use fact that any real number ($\mathbb{R}$) can be constructed as the limit of a sequence of rational #'s (Q)

Prob. Function satisfies AoP

$\boxed{1)\ P(A) \geq 0 \ \forall A \in \mathcal{B}}$ $\boxed{2)\ P(S) = 1}$

$\boxed{3)\ A_i \cap A_j = \emptyset \ \forall i \neq j \Rightarrow P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} A_i}$

Assume finite additivity: $A, B \in \mathcal{B}$

$\boxed{A \cap B = \emptyset \Rightarrow P(A \cup B) = P(A) + P(B)}$

Thm. $\forall A \in \mathcal{B}$, $P(A) := \sum_{\{i : s_i \in A\}} p_i$

$S = \{s_1, \dots, s_n\}$ - finite
or $S = \{s_1, s_2, \dots\}$ - countable

- physical reality (context) of the experiment motivates prob. assignment

1.2.2: Calculus of Prob.'s

use AoP to build properties $A, B \in \mathcal{B}$, P: prob function

$\boxed{P(\emptyset) = 0}$ $\boxed{P(A) \leq 1}$ $\boxed{P(A^c) = 1 - P(A)}$ $\boxed{P(A \cap B) \geq P(A) + P(B) - 1}$

$\boxed{P(B \cap A^c) = P(B) - P(A \cap B)}$ Bonferroni's inequality: bound simultaneous event (∩) in terms of individual events w.r.t. prob; only useful if ind. prob. events are sufficiently large

$\boxed{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$

$\boxed{A \subset B \Rightarrow P(A) \leq P(B)}$

∀ partition $C_1, C_2, \dots$

$\boxed{P(A) = \sum_{i=1}^{\infty} P(A \cap C_i)}$

(★) $\boxed{P\left(\bigcap_{i=1}^{\infty} A_i\right) \geq \sum_{i=1}^{\infty} P(A_i) - (n-1)}$

$\boxed{P\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} P(A_i)}$ (#)

(#) Boole's inequality is a more general version of (★) - Bonferroni's

1.2.3: Counting

FTOC: k separate tasks, i^{th} of which done in n_i ways; $i = 1, \dots, k$ then the entire job can be done in $n_1 \times \dots \times n_k = n^k$ ways

$n! := n \times (n-1) \times (n-2) \times \dots \times 3 \times 2 \times 1$, $0! = 1$
- # distinct ways the original n objects can be shuffled

Number of possible arrangements of size r from n objects

	w/o replacement	w/ replacement
ordered	$P_r^n = \frac{n!}{(n-r)!}$	n^r (FTOC)
unordered	$C_r^n = \frac{n!}{r!(n-r)!}$	$\binom{n+r-1}{r} = \frac{(n+r-1)!}{r!(n-1)!}$

$P_r^n = n(n-1)\cdots(n-r+1) = \frac{n!}{(n-r)!}$
- ordered arrangement of r distinct objects out of n total

$C_r^n = \binom{n}{r} = \frac{P_r^n}{r!} = \frac{n!}{r!(n-r)!}$
- # unordered combos of n objects chosen r at a time

Notes: • $\binom{n}{r}$ divides by redundant orderings
- $C_r^n \leq P_r^n$ since $r! \geq 1$

1.2.4: Enumerating options

use (1.2.3) techniques for finite S and if all outcomes equally likely
↳ then P(events) found via counting # outcomes in the event (classical model)

Prop. $S = \{s_1, \dots, s_n\}$. $P(\{s_i\}) = \frac{1}{N} \ \forall s_i \in S$.

$\Rightarrow \boxed{P(A) = \sum_{s_i \in A} P(\{s_i\}) = \sum_{s_i \in A} \frac{1}{N} = \frac{\# A \text{ elements}}{\# N \text{ elements}}}$

Inclusion/Exclusion: For n events $A_1, A_2, \dots, A_n$ (★)

$\boxed{P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{k=1}^{n} (-1)^{k+1} \left(\sum_{1 \leq i_1 < i_2 < \dots < i_k \leq n} P(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}) \right)}$

$\boxed{P(\text{at least one } A_i) = P\left(\bigcup_{i=1}^{n} A_i\right) = 1 - P\left(\bigcap_{i=1}^{n} A_i^c\right)}$

1.3: Conditional Prob & Independence

$P(A \mid B) := \frac{P(A \cap B)}{P(B)}$

(⇒) Joint = marginal × cond. ⇒ $\boxed{P(A \cap B) = P(B)P(A \mid B)}$

$\boxed{P(A_1 \cap \dots \cap A_n) = \prod_{i=1}^{n} P(A_i \mid A_1 \cap \dots \cap A_{i-1})}$

$\boxed{P(A \cap B) = P(A)P(B \mid A) = P(B)P(A \mid B)}$

Bayes' Formula $\boxed{P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}}$

$\mathbb{1}$ ind: $\boxed{\text{Ind} := P(A \cap B) = P(A)P(B)}$

$\boxed{A \perp\!\!\!\perp B \Leftrightarrow A \perp\!\!\!\perp B^c, A^c \perp\!\!\!\perp B, A^c \perp\!\!\!\perp B^c}$

Bayes' Thm. $\boxed{S = \bigcup_{i=1}^{\infty} A_i, P(A_i \mid B) = \frac{P(B \mid A_i)P(A_i)}{\sum_i P(B \mid A_i)P(A_i)} (*)}$

$\boxed{\text{mutual Ind}: P(A_1 \cap \dots \cap A_{i_k}) = \prod_{j=1}^{n} P(A_{i_j})}$

- turning around condit. prob's; (*) Law of Total Prob
- mutual ind $\Rightarrow$ ind

$\boxed{\forall i \ P(A_i \mid B) \propto P(B \mid A_i) P(A_i)}$

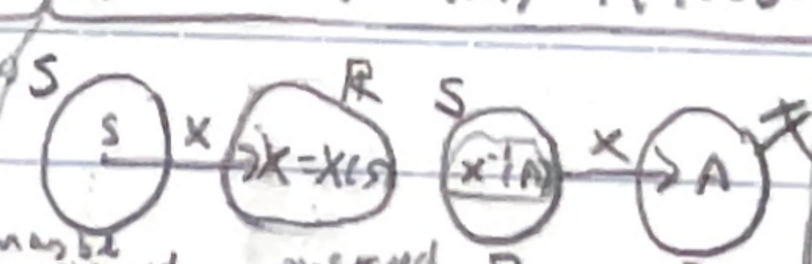
$P(A_i \mid B)$ = Bayesian Posterior Dist. (obs data); $P(B \mid A_i)$ = likelihood; $P(A_i)$ = Prior Dist.

1.4: Random variables

RV: function mapping from $X : S \to \mathbb{R}$
- easier to deal w/ summary var than original prob. structure

$S = \{s_1, \dots, s_n\}$, $\mathcal{X} = \{x_1, \dots, x_m\}$ $\boxed{P_X(X = x_i) = P(\{s_j \in S : X(s_j) = x_i\})}$

$\boxed{P_X(X \in A) = P(X^{-1}(A)) = P(\{s \in S : X(s) \in A\})}$ $\mathcal{X}$: uncountable range



$\boxed{f_X(x) = P(X = x) \ \forall x}$

$\boxed{F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt \ \forall x}$

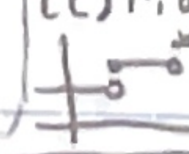
1.5: Distribution Functions

CDF: $F_X(x) = P(X \leq x) \ \forall x \in \mathcal{X}$

CDF Properties (a) $\lim_{x\to-\infty} F(x) = 0$, $\lim_{x\to\infty} F(x) = 1$ (b) nondecreasing function of x (c) right continuous, i.e. $\lim_{x \downarrow x_0} F(x) = F(x_0)$

$F'(x) > 0 \Rightarrow$ increasing

$\lim_{\epsilon \downarrow 0} F_X(x+\epsilon) = F_X(x)$



- can construct new RV from known constituent RV's

cont RV ⇔ continuous F_X
Discrete RV ⇔ Discrete F_X

$\boxed{X \overset{d}{=} Y \Leftrightarrow F_X(a) = F_Y(a) \ \forall a, b \in \mathbb{R}}$

$\boxed{X \overset{d}{=} Y := P(X \in A) = P(X \in B) \ \forall A, B \in \mathcal{B}'}$

$\mathcal{B}'$: smallest σ-algebra containing $(a,b), [a,b), (a,b], [a,b]$

1.6: Density & Mass Functions

- $X \sim F_X$ or $X \sim f_X$ contain all info abt dist. of RV X

$\boxed{f_X(x) = \frac{d}{dx} F_X(x)}$

$\boxed{P(X \leq x) = F_X(x)}$ $\boxed{X\text{-cont.}: P(a < X < b) = \dots = P(a \leq X \leq b)}$

$\boxed{X\text{-cont.} \Rightarrow P(X = x) = 0}$ PDF/PMF satisfies:

$\boxed{a)\ f_X(x) \geq 0 \ \forall x}$ $\boxed{b)\ \sum_x f(x) = 1, \int_{-\infty}^{\infty} f_X(x)\,dx = 1}$

$X \sim \text{Binomial}(n,p)$; $f(x) = \binom{n}{x} p^x (1-p)^{n-x}$, $x = 0, 1, \ldots, n$; $E(X) = np$, $V(X) = np(1-p)$, $M_X(t) = [(1-p) + pe^t]^n$

$X \sim \text{Normal}(\mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$; $\sigma > 0$; $E(X) = \mu$, $V(X) = \sigma^2$, $M_X(t) = \exp(\mu t + \frac{1}{2}\sigma^2 t^2)$

$X \sim N(0,1)$; $f = \frac{1}{\sqrt{2\pi}} \exp(-\frac{z^2}{2})$; $\chi^2_{df} = \sum_{i=1}^{df} Z_i^2$, $Z_i \sim N(0,1)$; $X \sim \text{Bernoulli}(p)$; $f = p^x (1-p)^{1-x}$, $x = 0, 1$, $p \in (0,1)$; $E(X) = p$, $V(X) = p(1-p)$, $M_X(t) = (1-p) + pe^t$

$X \sim \text{Unif}(a,b)$; $f = \frac{1}{b-a}$, $a < x < b$

$X \sim \text{Exp}(\theta)$, $f = \frac{1}{\theta} \exp(-\frac{x}{\theta})$, $x \geq 0$, $\theta > 0$; $E(X) = \theta$, $V(X) = \theta^2$; $M_X(t) = (1-\theta t)^{-1}$, $t < 1/\theta$

$X \sim \text{Gamma}(\alpha, \beta) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} \exp(-\frac{x}{\beta})$, $x > 0$, $\alpha, \beta > 0$; $E(X) = \alpha\beta$, $V(X) = \alpha\beta^2$, $M_X(t) = \left(\frac{1}{1-\beta t}\right)^\alpha$, $t < 1/\beta$; $X/\beta \sim \text{Gamma}(\alpha, 1)$

$X \sim \text{Cauchy}(\theta, \sigma) = \frac{1}{\pi\sigma}\left(1 + \left(\frac{x-\theta}{\sigma}\right)^2\right)^{-1}$; $\sigma > 0$; $x \in \mathbb{R}$; $E(X)$, $V(X)$, $M_X(t)$ DNE; $X \sim \text{NegBin}(r,p) = \binom{r+x-1}{x} p^r (1-p)^x$; $p \in (0,1)$

$X \sim \text{Cauchy}(0,1) = \frac{1}{\pi}(1+x^2)^{-1}$

IBP: $\int u\,dv = uv - \int v\,du$; choose u: LIATE; $\frac{d}{dx}$ ... s.t. integral is ...

$|x| \leq y = \begin{cases} x \leq y \\ x \geq -y \end{cases}$; $|x| \geq y = \begin{cases} x \geq y \\ x \leq -y \end{cases}$

$\frac{d}{d\theta} \tan(\theta) = \sec^2(\theta) = \frac{1}{\cos^2(\theta)}$; $\ln(AB) = \ln(A) + \ln(B)$

u-sub: $\int \frac{x}{1+x^2} dx$ ...

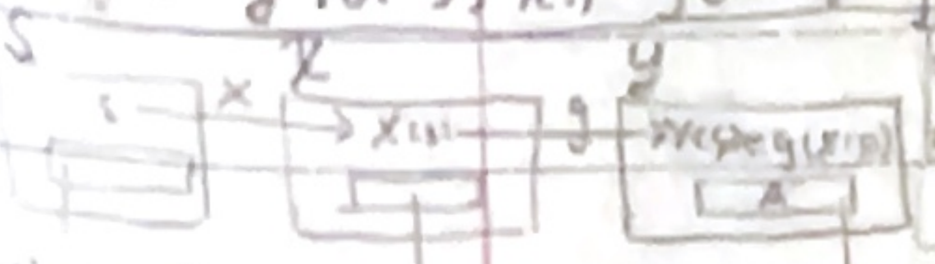
$\int_{-\infty}^{\infty} f(|x-a|)dx = \int_{-\infty}^{a} f(a-x)dx + \int_{a}^{\infty} f(x-a)dx$

$tx - \frac{x^2}{2} = -\frac{1}{2}(x^2 - 2tx) = -\frac{1}{2}((x-t)^2 - t^2) = -\frac{(x-t)^2}{2} + \frac{t^2}{2}$

2.1: Dist. of RV Functions

$Y = g(X)$ new RV

Let $g: \mathcal{X} \to \mathcal{Y}$ for S; $Y(s) = g(X(s)) = g \circ X(s)$



$Y^{-1}(A) = X^{-1}(g^{-1}(A))$; $g^{-1}(A) = \{x : g(x) \in A\}$; $P_Y(A)$

$g^{-1}(y) = \{x : g(x) = y\}$ is singleton set

Discrete X; $Y = g(X)$; $\mathcal{Y} = \{y : y = g(x), x \in \mathcal{X}\}$

$f_Y(y) = P(Y=y) = \sum_{x \in g^{-1}(y)} P(X=x) = \begin{cases} \sum_{x \in g^{-1}(y)} f_X(x), & y \in \mathcal{Y} \\ f(y) = 0, & y \notin \mathcal{Y} \end{cases}$

e.g. (Binom): $Y = g(X) = n - X$; $X \sim \text{Bin}(n,p)$; $\binom{n}{y} = \binom{n}{n-y}$ $\Rightarrow Y \sim \text{Bin}(n, 1-p)$

cont. X; $Y = g(X)$; $F_Y(y) = P(Y \leq y) = P(g(X) \leq y) = P(\{x \in \mathcal{X} : g(x) \leq y\}) = \int f_X(x)\,dx$

e.g. (Square): $\{x : g(x) \leq y\}$; $F_Y(y) = P(Y \leq y) = P(X^2 \leq y)$; $X \sim N(0,1)$; $Y = g(X) = X^2 \sim \chi^2_1$; $= P(|X| \leq \sqrt{y})$, $y \in \mathbb{R}$

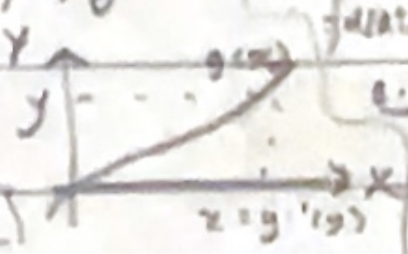
$F_X(\sqrt{y}) - F_X(-\sqrt{y})$; $f(y) = \frac{d}{dy} F_Y(y) = \ldots = \frac{1}{\sqrt{2\pi}} \exp(-\frac{y}{2}) \sim \chi^2_1$

Def (monotonic): $g\uparrow$, i.e. $g(x) > g(y)$ if $x > y$; $g\downarrow$, i.e. $g(x) < g(y)$ if $x > y$

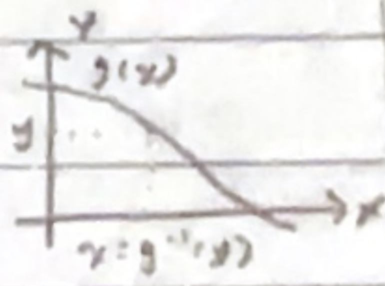
Thm. X has $F_X(x)$; $Y = g(X)$; $g: \mathcal{X} \to \mathcal{Y}$

- $g\uparrow$: $\{x : g(x) \leq y\} = \{x \leq g^{-1}(y)\}$

$F_Y(y) = \int f_X(x)\,dx = F_X(g^{-1}(y))$



- $g\downarrow$: $\{x : g(x) \leq y\} = \{x \geq g^{-1}(y)\}$; $F_Y(y) = \int f_X\,dx = 1 - F_X(g^{-1}(y))$



a) $f_Y(y) = f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y)$; b) $f_Y(y) = -f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y)$

change-of-var: $f_Y(y) = f_X(g^{-1}(y)) \left|\frac{d}{dy} g^{-1}(y)\right|$

e.g. $X \sim \text{Unif}(0,1)$, $Y = g(X) = -\log(X)$; $g^{-1}(y) = x = e^{-y}$; $g\downarrow$ since $g'(x) = -1/x < 0$ for $x \in (0,1)$

$F_X(x) = \int_0^x 1\,dx = x$; $F_Y(y) = 1 - F_X(g^{-1}(y)) = 1 - e^{-y}$

e.g. $X \sim \text{Gamma}(n, \beta)$, $Y = g(X) = \frac{1}{X}$; $g^{-1}(y) = \frac{1}{y} = x$; $g\downarrow$ on $\mathcal{X}$ and $\frac{d}{dy} g^{-1}(y) = -1/y^2$, using (a)

$f_Y(y) = \frac{1}{\Gamma(n)\beta^n} \left(\frac{1}{y}\right)^{n-1} e^{-1/(\beta y)} \frac{1}{y^2} = \frac{1}{\Gamma(n)\beta^n} y^{-n-1} e^{-1/(\beta y)}$

PIT: X cont., $Y = F_X(X) = g(X) \Rightarrow Y \sim \text{Unif}(0,1)$

$y \in (0,1)$, $P(Y \leq y) = P(F_X(X) \leq y)$, $F_X\uparrow$

$= P(F_X^{-1}[F_X(X)] \leq F_X^{-1}(y)) = P(X \leq F_X^{-1}(y))$

$= F_X(F_X^{-1}(y))$, F_X cont. $= y$; $P(Y \geq 1) = 1$, $P(Y \leq 0) = 0$

$F_X^{-1}(y)$: $P(X^{-1} \leq F_X^{-1}(y)) = P(X \leq \ldots)$

Inf(A): $\begin{cases} a \in \mathbb{R} : a \leq x \;\forall x \in A \\ -\infty, \text{ DNE} \end{cases}$ e.g. $\inf(a, \infty) = a$

$F_X^{-1}(y) = \inf\{x : F_X(x) \geq y\}$ $\forall y \in (0,1)$

- $F_X^{-1}(1) = +\infty$, $F_X^{-1}(0) = -\infty$

Thm. If $U \sim \text{Unif}(0,1)$, $Z = F_X^{-1}(U)$ ... $\inf\{x : F_X(x) \geq y\}$ $\forall y \in (0,1)$; then $F_X^{-1}(y) = F_X(x)$

- generating RVs from unif RV, no cont. restriction

e.g. $X \in \{a, b\}$; $P(X=a) = p$, $P(X=b) = 1-p$, $a < b$, $Y \sim \text{Unif}(0,1)$

$F_X^{-1}(y) = \begin{cases} -\infty, & y = 0 \\ a, & 0 < y \leq p \\ b, & p < y \leq 1 \end{cases}$ $= \inf\{x : F_X(x) \geq y\}$ $\forall y \in (0,1)$

$P(F_X^{-1}(Y) = a) = P(0 < Y \leq p) = p$

$P(F_X^{-1}(Y) = b) = P(p < Y \leq 1) = 1-p$

$P(F_X^{-1}(Y) = \ldots) = P(Y = 0) = 0$

2.2: Expectation

$E[g(X)] = \begin{cases} \int_{-\infty}^{\infty} g(x) f_X(x)\,dx, & X \text{ cont.} \\ \sum_{x \in \mathcal{X}} g(x) f_X(x), & X \text{ disc.} \end{cases}$

- If $x \in \mathcal{X} \notin \mathbb{R}$, then $E(X)$ may be undefined, while $E(g(X))$ may be defined
- Proper exp. $\exists$ i.f.f. $\left(\int |x| f_X(x)\,dx < \infty\right)$, i.e. integral finite for ...
- $E(X)$ DNE if $\int x f_X(x)\,dx$ not abs. convergent

$E(X) = \begin{cases} \int x \cdot f_X(x)\,dx, & X \text{ cont.} \\ \sum x \cdot f_X(x), & X \text{ disc.} \end{cases}$

Prop. X RV $\sim P_X$ for $X_1, \ldots, X_n$ n-iid of P_X, then $Y \sim P_n = \{x_1, \ldots, x_n\}$, $y \in \mathcal{Y} = \{x_1, \ldots, x_n\}$ and $P(Y = x_i) = \frac{1}{n}$ for $i = 1, \ldots, n$

$\Rightarrow E(Y) = \frac{1}{n}(X_1 + \cdots + X_n) \neq E(X)$

avg. of data sample $X_1, \ldots, X_n$

e.g. (Exp): $X \sim \text{Exp}(\lambda)$, $\int f(x)\,dx = 1$; $E(X) = \int x f(x)\,dx = \lambda$

e.g. (Bin): $X \sim \text{Bin}(n,p)$; $\sum f(x) = 1$ ... $(x+a)^n = \sum_{k=0}^{n} \binom{n}{k} x^k a^{n-k}$ $= (p + (1-p))^n = 1$

$E(X) = \sum x f(x)$; $x\binom{n}{x} = n\binom{n-1}{x-1}$, let $y = x-1$ ... $= np(p + 1-p)^{n-1} = np$.

e.g. (Cauchy): $X \sim \text{Cauchy}(0,1)$, $E|X| = \int_{-\infty}^{\infty} \frac{|x|}{\pi(1+x^2)}\,dx = 2\int_0^\infty \frac{x}{\pi(1+x^2)}\,dx = \lim \frac{1}{\pi} \log(1+x^2) = \infty$

$\int f(x)\,dx = \ldots$ change of var $\arctan(x)$ ... $\frac{1}{\pi}(\frac{\pi}{2} + \frac{\pi}{2}) = 1$; $E(X) = \int x f(x)\,dx$ ... DNE

Linearity: $E(aX + bY + c)$ for $a, b, c \in \mathbb{R}$

$= aE(X) + bE(Y) + c$; $X = g_1(Z)$, $Y = g_2(Z)$

Pf. X, Y RV $\in \mathbb{R}$. $E(aX + bY + c) = \int (a g_1(z) + b g_2(z) + c) f(z)$

$= aE(g_1(Z)) + bE(g_2(Z)) + c$ $\square$

e.g. Let $X \sim \text{Bin}(n,p)$; $Y_i \sim \text{Bernoulli}(p)$

$E(Y_i) = \sum y f(y) = 1(p) + 0(1-p) = p$

$E(X) = E(Y_1) + \cdots + E(Y_n) = np$

Prop. i) $X \geq 0$ a.s. $(P(X \geq 0) = 1) \Rightarrow E(X) \geq 0$

ii) $X \geq Y$ a.s. $(P(X \geq Y) = 1) \Rightarrow E(X) \geq E(Y)$

iii) $P(X \in [a,b]) = 1 \Rightarrow E(X) \in [a,b]$

Prop. Let $X_1, \ldots, X_n$ n-iid, $X = X_1 + \cdots + X_n$

$V(X) = V(X_1) + \cdots + V(X_n) = n \cdot Var(X_1)$

MGF: $M_X(t) = E(e^{tX})$ provided $t \in$ some $(-h, h)$ for $h > 0$

- can characterize dist.
- uniquely determines all moments

$E(X) = \frac{d}{dt} \int e^{tx} f(x)\,dx$ ... $\int x e^{tx} f(x)|_{t=0} = \int x f(x)\,dx$, $\forall n \in \mathbb{N}$, $E(X^n) = \frac{d^n}{dt^n} M_X(t)\Big|_{t=0} = M_X^{(n)}(0)$

- gives meaning of moment generating

e.g. (Gamma): $\Gamma(\alpha) = \frac{1}{\beta^\alpha} \int x^{\alpha-1} e^{-x/\beta}\,dx$

letting $y = x/\beta$, $x = \beta y \Rightarrow dx = \beta\,dy$; $= \int_0^\infty y^{\alpha-1} e^{-y}\,dy$

$\int f(x)\,dx = \frac{\Gamma(\alpha)}{\Gamma(\alpha)} = 1$, $M_X(t) = \int e^{tx} f(x)$

$= \frac{1}{\Gamma(\alpha)\beta^\alpha} \int_0^\infty x^{\alpha-1} \exp\left(\frac{(t\beta - 1)x}{\beta}\right) = \left(\frac{1}{1-\beta t}\right)^\alpha$, $t < 1/\beta$

$t = \frac{1}{\beta} \Rightarrow \frac{1}{\beta} \cdot t \leq 0 \Rightarrow M_X(t) = +\infty$

$E(X) = \alpha(1-\beta t)^{-\alpha-1} \beta|_{t=0} = \alpha\beta$

e.g. (Binomial): $X \sim \text{Bin}(n,p)$

$M_X(t) = E(e^{tX}) = \sum e^{tx} f(x)$

binom. theorem $= (e^t p + (1-p))^n$

$E(X^k) = M_X^{(k)}(0)|_{t=0}$.

Thm. $\forall a, b \in \mathbb{R}$,

$M_{aX+b} = E(e^{t(aX+b)}) = e^{bt} E(e^{taX}) = e^{bt} M_X(at)$

e.g. $M_X(t) = p^r/(1-(1-p)e^t)^r$. Let $Y = 2pX$;

$M_Y(t) = M_{2pX}(t) = M_X(2pt) = p^r/(1-(1-p)e^{2pt})^r$

2.3: Moments

n-th moment: $\mu'_n := E(X^n)$; n-th central mom.: $\mu_n := E(X - E(X))^n$, $\forall n \in \mathbb{N}$

$V(X) = E(X - E(X))^2 = E(X^2) - 2E^2(X) + E^2(X) = E(X^2) - [E(X)]^2$; $SD(X) := \sqrt{Var(X)}$

$V(aX + b) = E[(aX+b)^2 - (aE(X) + b)^2] = \ldots = a^2(E(X^2) - E^2(X)) = a^2 Var(X)$

e.g. (Exp): $X \sim \text{Exp}(\lambda)$, $V(X) = E(X^2) - E^2(X)$ where $E(X^2) = \int x^2 f(x)$ = IBP $2\lambda^2$ $\Rightarrow V(X) = 2\lambda^2 - \lambda^2 = \lambda^2$

e.g. (Bin): $X \sim \text{Bin}(n,p)$; $E(X^2) = \sum x^2 f(x)$, $x\binom{n}{x} = n\binom{n-1}{x-1}$, $= \sum xn\binom{n-1}{x-1} p^x (1-p)^{n-x}$;

$y = x-1 \Rightarrow x = y+1$ ... expand w/ $n x(x+a)^n = \sum_{k=0}^{n} k\binom{n}{k} x^k a^{n-k}$

$= np[(n-1)p(p + (1-p))^{n-2} + (p + (1-p))^{n-1}] = n(n-1)p^2 + np$

$\Rightarrow V(X) = E(X^2) - E^2(X) = n(n-1)p^2 + np - (np)^2 = np - np^2 = np(1-p)$

$\Leftarrow$ $Y_i \sim \text{Bernoulli}(p)$, $V(X) = V(Y_1) + \cdots + V(Y_n) = n\,Var(Y_1) = np(1-p)$

Thm. Let $F_X(x)$, $F_Y(y)$ valid cdfs; i) if X and Y have bounded support $\{x : f(x) > 0\}$, $F_X(u) = F_Y(u)$ $\forall u$ iff $E(X^r) = E(Y^r)$ $\forall r = 0, 1, \ldots$

ii) $M_X(t)$ and $M_Y(t)$ exist, $M_X(t) = M_Y(t)$, $t \in (-h, h) \Rightarrow F_X(u) = F_Y(u)$ $\forall u$

2.3: Moments (cont.)

Def: $X_n \xrightarrow{d} X$ i.f.f. $F_{X_n}(x) \to F_X(x)$ at all continuity points of F_X

Thm. Let $X_1, \ldots, X_n$ RVs with $M_{X_i}(t)$

$\lim_{i \to \infty} M_{X_i}(t) \to M_X(t) \Rightarrow X_i \xrightarrow{d} X$

$\exists h > 0$ s.t. $\forall t \in (-h, h)$, $M_{X_i}(t) = M_X(t)$

- moments do not capture all info. about dist. in unbounded setting, e.g. $X \sim f_1$, $Y \sim f_2$ unbounded

$Z \sim N(0,1)$. Let $X = e^Z \Rightarrow Z = g^{-1}(X) = \ln(X)$

$f(x) = f_Z(g^{-1}(x)) \left|\frac{d}{dx} g^{-1}(x)\right| \sim \text{LogNorm}(0,1)$

$F_2 = f_1(x)(1 + \sin(2\pi(\log x)))$

where $E(X^r) = E(Y^r)$, $r = 0, 1, \ldots$ but $F(x) \neq F(y)$.

e.g. (Poisson Approx.): $X_n \sim \text{Bin}(n,p)$, $Y \sim \text{Pois}(\lambda)$, assume $\begin{cases} np = \lambda \text{ constant} \\ n \to \infty, p \to 0 \end{cases}$

$M_{X_n}(t) = (pe^t + (1-p))^n$; using $p = \frac{1}{n}p$, $np = \lambda$

$\Rightarrow \lim_{n \to \infty} (1 + \frac{\lambda}{n}(e^t - 1))^n = \exp(\lambda(e^t - 1)) = M_Y(t) \Rightarrow F_{X_n}(x) \to F_Y(x)$ as $n \to \infty$ $\forall x$ where F_Y is cont., i.e. $x \in \mathbb{R}$

2.4: Tools

- $f(x, \theta)$, $a(\theta)$, $b(\theta)$ differentiable wrt θ

Thm (Leibniz's): $f(x, \theta)$ cont. on $x \in [x_0, x_1]$

$\frac{d}{d\theta} \int_{a(\theta)}^{b(\theta)} f(x, \theta)\,dx = f(b(\theta), \theta) \frac{d}{d\theta} b(\theta) - f(a(\theta), \theta) \frac{d}{d\theta} a(\theta) + \int_{a(\theta)}^{b(\theta)} \frac{\partial}{\partial\theta} f(x, \theta)\,dx$

Let $a(\theta) = a$, $b(\theta) = b$; $\frac{d}{d\theta} \int_a^b f(x, \theta)\,dx = \int_a^b \frac{\partial f}{\partial\theta}(x, \theta)\,dx$

Thm. - $h(x, y)$ cont. at $y = y_0$ for each fixed x - there is a function $g(x)$ s.t. $\begin{cases} |h(x,y)| \leq g(x) \;\forall x, y \\ \int_{-\infty}^{\infty} g(x)\,dx < \infty \end{cases}$ (converge cond.)

$\lim_{y \to y_0} \int_{-\infty}^{\infty} h(x, y)\,dx = \int_{-\infty}^{\infty} \lim_{y \to y_0} h(x, y)\,dx$

Pointwise conv.: $\lim_{n \to \infty} \sum_{x=0}^{n} h(x, \theta) =: S_n(\theta) = \sum_{x=0}^{\infty} h(x, \theta) =: S(\theta)$

Sup A: smallest $s \in \mathbb{R}$ s.t. $s \geq a$ $\forall a \in A$, e.g. $\sup(a, r) = r$.

uniform conv.: $\sup_{\theta \in (a,b)} |S_n(\theta) - S(\theta)| \to 0$ as $n \to \infty$

Thm. - $\sum_{x=0}^{\infty} h(x, \theta)$ exists (converges pointwise) $\forall \theta \in (a,b)$

1) $\frac{\partial}{\partial\theta} h(x, \theta)$ is cont. in θ for each x 2) $\sum_{x=0}^{\infty} \frac{\partial}{\partial\theta} h(x, \theta)$ converges uniformly $\forall \theta$ in closed sub-int of (a,b)

$\frac{d}{d\theta} \sum_{x=0}^{\infty} h(x, \theta) = \sum_{x=0}^{\infty} \frac{\partial}{\partial\theta} h(x, \theta)$; $X \sim \text{Geo}(\theta)$, $f = \theta(1-\theta)^x$, $x = 1, 2, \ldots$, $\theta \in (0,1)$

e.g. $X \sim \text{Geo}(\theta)$; $\frac{d}{d\theta} \sum f = \sum \frac{\partial f}{\partial\theta} \Rightarrow h(x, \theta) = \theta(1-\theta)^x$

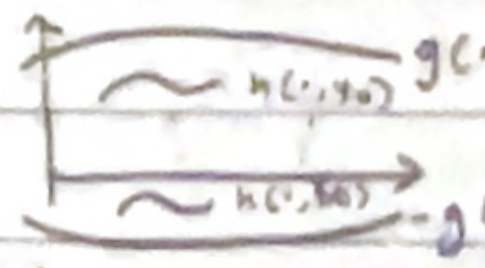
$\frac{\partial}{\partial\theta} h(x, \theta) = (1-\theta)^x - \theta x(1-\theta)^{x-1}$ cont. in θ

$S_k(\theta) = \sum_{x=0}^{k} \frac{\partial}{\partial\theta} h(x, \theta) = (k+1)(1-\theta)^k \to 0$ as $k \to \infty$ (PC)

$\sup_{\theta \in [c,d]} |S_k(\theta) - 0| \to 0$ as $k \to \infty$ $\forall [c,d] \subset (0,1)$ (UC)

Thm. - for each x, $f(x, \theta)$ is differentiable wrt θ, $\theta \in \Theta$

- For each $\theta \in \Theta$, there is a function $g(\cdot, \theta)$ and $\delta > 0$ s.t.



$\begin{cases} \left|\frac{\partial}{\partial\theta} f(x, \theta)\right|_{\theta = \theta'} \leq g(x, \theta) \;\forall \theta' \in (\theta - \delta_0, \theta + \delta_0) \\ \int_{-\infty}^{\infty} g(x, \theta)\,dx < \infty \text{ for each } \theta \in \Theta \end{cases}$

then $\frac{d}{d\theta} \int_{-\infty}^{\infty} f(x, \theta)\,dx = \int_{-\infty}^{\infty} \frac{\partial}{\partial\theta} f(x, \theta)\,dx$ $\forall \theta \in \Theta$

e.g. (i): $X \sim \text{Exp}(\lambda)$, $\frac{d}{d\lambda} E(X^n) = \frac{d}{d\lambda} \int_0^\infty \ldots = \int_0^\infty \frac{\partial}{\partial\lambda} \ldots$; $\left|\frac{\partial}{\partial\lambda} h(x, \lambda)\right| = x^n e^{-x/\lambda} \ldots \left|\frac{x}{\lambda} - 1\right| \leq x^n e^{-x/(\lambda + \delta_0)} \ldots = g(x, \lambda)$, $x > 0$, $\lambda > \delta_0$

$X \sim \text{LogNormal}(\mu, \sigma^2)$; $f = \frac{1}{\sqrt{2\pi}\sigma} \frac{1}{x} \exp\left(-\frac{(\log(x) - \mu)^2}{2\sigma^2}\right)$, $x > 0$, $\sigma > 0$; $E(X) = \exp(\mu + \frac{\sigma^2}{2})$, $V(X) = \exp(2(\mu + \sigma^2)) - \exp(2\mu + \sigma^2)$; $E(X^r) = \exp(r\mu + \frac{r^2\sigma^2}{2})$

$X \sim \text{Poisson}(\lambda)$; $f = (e^{-\lambda} \lambda^x)/x!$, $\lambda \geq 0$, $x \geq 0$; $E(X) = \lambda = V(X)$, $M_X(t) = e^{\lambda(e^t - 1)}$

$\lim_{x \to \infty} (1 + \frac{a}{x})^x = e^a$; L'Hôpital: $\lim_{x \to a} \frac{f(x)}{g(x)}$ ind. $= \lim_{x \to a} \frac{f'(x)}{g'(x)}$

Thm. Suppose $\sum_{x=0}^{\infty} h(x, \theta)$ converges pointwise $\forall \theta \in (a,b)$

1) $h(x, \theta)$ cont. in θ $\forall$ fixed x 2) $\sum_{x=0}^{\infty} h(x, \theta)$ converges uniformly on $[a,b]$. Then $\int_a^b \sum_{x=0}^{\infty} h(x, \theta)\,d\theta = \sum_{x=0}^{\infty} \int_a^b h(x, \theta)\,d\theta$

$\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$, $|r| < 1$

Pre-Test 1

$\bigcup_{i=1}^{\infty} A_i = \{x \in S \mid x \in A_i \text{ for some } i\}$

$\bigcap_{i=1}^{\infty} A_i = \{x \in S \mid x \in A_i \text{ for all } i\}$

$\mathcal{B}$: closed under complementation, $\bigcap^{\infty}$, $\bigcup^{\infty}$; show sets $\in \mathcal{B}$ via obtaining from known elements of $\mathcal{B}$ via countably many set ops.

	w/o replacement	w/ replacement
ordered	$P^n_r = \frac{n!}{(n-r)!}$	n^r
unordered	$C^n_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$	$\binom{n+r-1}{r} = \frac{(n+r-1)!}{r!(n-1)!}$

$P(\text{at least one } A_i) = P(\bigcup_{i=1}^{n} A_i) = 1 - P(\bigcap_{i=1}^{n} A_i^c)$

I.E. generalizes: $P(A \cup B \cup C) = P(A) + P(B) + P(C) - [P(A\cap B) + P(A \cap C) + P(B \cap C)] + P(A \cap B \cap C)$

Bayes: $P(A|B) = P(A\cap B)/P(B) \Leftrightarrow P(A \cap B) = P(A|B)P(B)$

chain: $P(A\cap B \cap C) = P(A)P(B\cap C|A) = P(A)P(B|A)P(C|B,A)$

Bayes: let $S = \bigcup_{i=1}^{\infty} A_i$, $P(B) = \sum_i P(B \cap A_i) = \sum_i P(B|A_i)P(A_i)$ → law of tot. prob.

$P(A_i|B) = [P(B|A_i)P(A_i)] / [\sum_j P(B|A_j)P(A_j)]$

RV: $P_X(X \in A) = P(X^{-1}(A)) = P(\{s \in S \mid X(s) \in A\})$

CDF: $F_X(x) = P(X \le x)\ \forall x \in \mathcal{X}$ $\begin{cases} \lim_{x\downarrow -\infty} = 0, \lim_{x \uparrow \infty} = 1 \\ \text{nondecreasing}, \lim_{x \downarrow x_0} F(x) = F(x_0) \end{cases}$

PDF: $F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt\ \forall x \in \mathcal{X} \Leftrightarrow f_X(x) = \frac{d}{dx}F_X(x)$

PMF: $f_X(x) = P(X = x)\ \forall x$; $\begin{cases} f(x) \ge 0\ \forall x \in \mathcal{X} \\ \sum_x f(x) = 1,\ \int f(x) = 1 \end{cases}$

Pre-Test 2

$Y(s) = g(X(s))$; $Y^{-1}(A) = X^{-1}(g^{-1}(A))$

Prop. Let X-cont, $Y = g(X)$; then $F(y) = P(Y \le y) = P(g(x) \le y) = P(x \in \mathcal{X} : g(x) \le y) = \int_{\{x: g(x) \le y\}} f(x)\,dx$.

Thm. $X: F_X(x)$, $\mathcal{X} = \{x \mid f(x) > 0\}$, $Y = g(X)$, then

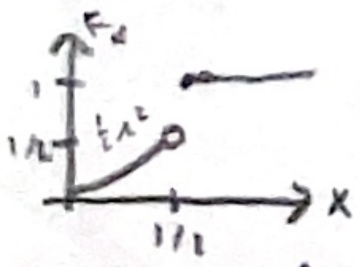
i) $g: \mathcal{X} \to \mathcal{Y}\ \uparrow$, then $F_Y(y) = F_X(g^{-1}(y))\ \forall y \in \mathcal{Y}$

ii) $g: \mathcal{X} \to \mathcal{Y}\ \downarrow$, then $F_Y(y) = 1 - F_X(g^{-1}(y))\ \forall y \in \mathcal{Y}$

cov: $f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$

(min. set equivalent)

Inf: Let $A \subset \mathbb{R}$. $\inf A :=$ largest $a \in \mathbb{R} \cup \{-\infty\}$ s.t. $a \le x\ \forall x \in A$, e.g. $\inf((a,b]) = a \ne \min(a,b)$ DNE

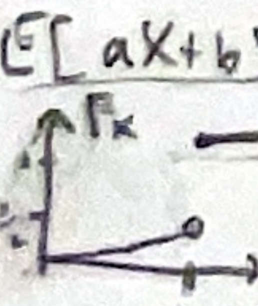


$F^{-1}(p) = \inf\{x \in \mathbb{R} : F(x) \ge p\} = \begin{cases} -\infty, & p < 0 \\ \sqrt{2p}, & 0 \le p < 1/2 \\ 1, & 1/2 \le p \le 1 \\ \infty, & p > 1 \end{cases}$

$F_X^{-1}(1) = +\infty$, $F_X^{-1}(0) = -\infty$ $\forall p \in (0,1)$

$E[g(X)] = \begin{cases} \int_{\mathbb{R}} g(x)f(x)\,dx \\ \sum_{x \in \mathcal{X}} g(x)f(x)\,dx \end{cases}$ e.g. $g(x) = x \Rightarrow \begin{cases} \int x f(x)\,dx \\ \sum x f(x) \end{cases}$

$E[aX + bY + c] = aE(X) + bE(Y) + c$, $a, b, c \in \mathbb{R}$: const.



$F_X(x) = \begin{cases} 0, & x < 0 \\ \frac{1}{2}x^2, & x \in [0,1) \\ 1, & x \ge 1 \end{cases}$ $f(x) = x, x \in [0,1)$; $P(X=1) = \Delta$: $F(1) - F(1^-) = \frac{1}{2}$; $E(X) = \int_0^1 x f(x)\,dx + x_0 \cdot \Delta$

assign Δ mass @ discontinuity point x_0.

$\mu'_n := E(X^n)$; $\mu_n := E(X - E(X))^n$; $V(X) = E(X - E(X))^2$

$V(X) = E(X^2) - [E(X)]^2$; $V(aX + b) = a^2 \mathrm{var}(X)$

Thm. Let $X_i \overset{iid}{\sim}$, $X = X_1 + \dots + X_n$; $\mathrm{var}(X) = n \cdot \mathrm{var}(X_i)$

mgf: $M_X(t) = E(e^{tX})$; provided $E(e^{tX})$ ∃ in some nghbd of $t = 0$, i.e. $t \in (-h, h)$ for $h > 0$

characterize the dist.

• justify $\int_{\mathbb{R}} e^{tx} f(x)\,dx$ via $|e^{tx} f(x)|$ dom. conv. thm.

Thm. $\forall n \in \mathbb{N}$, $EX^n = \frac{d^n}{dt^n} M_X(t)\big|_{t=0}$

Thm. $\forall a, b \in \mathbb{R}$: $M_{aX+b} = e^{bt} M_X(at)$

• Justify $M_X'(t) = \frac{d}{dt}\int_{\mathbb{R}} e^{tx} f(x)\,dx \overset{(*)}{=} \int_{\mathbb{R}} x e^{tx} f(x)\,dx$ via $|x e^{tx} f(x)|$ dom. conv. thm.

Thm. $F_X(x)$, $F_Y(y)$ - moments ∃. then $F_X(u) = F_Y(u)\ \forall u \Leftrightarrow E(X^r) = E(Y^r)\ \forall r = 0, 1, \dots$

$M_X(t) = M_Y(t)\ \forall t \in (-h, h)$ for $h > 0 \Leftrightarrow F_X(u) = F_Y(u)\ \forall u$

Thm. $\exists X_i$ whenever $F_{X_i}(x) \to F_X(x)\ \forall$ cont. point of F_X.

PT2

under conditions

• $\frac{d}{d\theta}\int_a^b f(x,\theta)\,dx = \int_a^b \frac{\partial f}{\partial \theta}(x,\theta)\,d\theta$

• $\lim_{y \to y_0} \int_{\mathbb{R}} h(x,y)\,dx = \int_{\mathbb{R}} \lim_{y \to y_0} h(x,y)\,dx$

• $\frac{d}{d\theta}\int_{\mathbb{R}} f(x,\theta)\,dx = \int_{\mathbb{R}} \frac{d}{d\theta} f(x,\theta)\,d\theta$

• $\frac{d}{d\theta}\sum_{x=0}^{\infty} h(x,\theta) = \sum_{x=0}^{\infty} \frac{\partial}{\partial\theta} h(x,\theta)$

• $\int_a^b \sum_{x=0}^{\infty} h(x,\theta)\,d\theta = \sum_{x=0}^{\infty} \int_a^b h(x,\theta)\,d\theta$

Thm. Suppose $h(x,y)$ cont. @ $y = y_0$ ∀ fixed x, then, there is a dominating (envelope) fx g s.t. $\begin{cases} |h(x,y)| \le g(x)\ \forall x, y \\ \int_{\mathbb{R}} g(x)\,dx < \infty \end{cases}$

Pointwise convergence: $\underbrace{\lim_{n\to\infty} \sum_{x=0}^{n} h(x,\theta)}_{S_n(\theta)} = \underbrace{\sum_{x=0}^{\infty} h(x,\theta)}_{S(\theta)}$

sup. Let $A \subset \mathbb{R}$; $\sup A :=$ smallest $s \in \mathbb{R}$ s.t. $s \ge a\ \forall a \in A$, e.g. $\sup(A = (u,v)) = v \ne \max(u,v)$

unif. conv: $\sup_{\theta \in (0,n)} |S_n(\theta) - S(\theta)| \to 0$ as $n \to \infty$

P2(0)

• odd: $f(-x) = -f(x)$, e.g. $x, x^3, \sin^3(x)$

• even: $f(-x) = f(x)$, e.g. $x^2, |x|, \cos(x)$

Thm. $\int_{-a}^{a} g(x)\,dx = 0$ if g odd, and $g(x)$ dist. symmetric abt zero

Prop. odd × odd = even; odd × even = odd; even × even = even

• $\int_{\mathbb{R}} f(|x - a|)\,dx = \int_{-\infty}^{a} \underbrace{f(a-x)}_{x<a}\,dx + \int_a^{\infty} \underbrace{f(x-a)}_{a<x}\,dx$

• computing the square: $tx - \frac{a}{2}x^2 = -\frac{a}{2}(x^2 - \frac{2t}{a}x) = -\frac{a}{2}(x - \frac{t}{a})^2 + \frac{t^2}{a}$

$\int_{\mathbb{R}} \exp(\frac{-a}{2}(x-m)^2)\,dx = \sqrt{\frac{2\pi}{a}}$ for $a > 0$

• $\int_{\mathbb{R}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)dx = \sqrt{2\pi\sigma^2}$

Ch 3: Dist Fun's

• $X \sim \mathrm{unif}(1,N)$; $p(x) = \frac{1}{N}, n = 1, \dots, N$; $E(X) = \frac{N+1}{2}$, $V(X) = [(N+1)(N-1)]/12 = (N-1)^2/12$

• $X \sim \mathrm{HypGeom}(N, M, K)$: N balls, M-red, $N-M$ G; pick K balls w/o replac; X: # red

$p(x) = \frac{1}{\binom{N}{K}} \cdot \binom{M}{x}\binom{N-M}{K-x}$, $x = 0, \dots, K$; $E(X) = \frac{KM}{N}$,

$V(X) = \frac{KM}{N} \cdot \frac{(N-M)(N-K)}{N(N-1)}$; $\sum_{x=0}^{K}\binom{M}{x}\binom{N-M}{K-x} = \binom{N}{K}$

• $X \sim \mathrm{Bernoulli}(p)$; $p(x) = \begin{cases} P(X=1|p) = p \\ P(X=0|p) = 1-p \end{cases}$ $\begin{cases} EX = p \\ VX = p(1-p) \end{cases}$

• Suppose $X_i \overset{iid}{\sim} \mathrm{Bernoulli}(p)$ $i = 1, \dots, n$; Y: # successes

$Y \sim \mathrm{Binomial}(n,p)$; $p(y) = \binom{n}{y}p^y(1-p)^{n-y}$; $\begin{cases} EY = np; VY = np(1-p) \\ M_Y(t) = (pe^t + 1 - p)^n \end{cases}$

$(x+y)^n = \sum_{k=0}^{n}\binom{n}{k}x^{n-k}y^k$, $x\binom{n}{x} = n\binom{n-1}{x-1}$

• $X \sim \mathrm{Poisson}(\lambda)$, $\lambda > 0$; $p(x) = e^{-\lambda} \cdot \frac{\lambda^x}{x!}$, $x = 0, 1, \dots$; $E(X) = \lambda = V(X)$

$e^{\lambda} = 1 + \lambda + \frac{\lambda^2}{2!} + \dots + \frac{\lambda^x}{x!} + \dots$; $M_X(t) = \exp(\lambda(e^t - 1))$

approx: $X_n \sim \mathrm{Bin}(n, p_n)$, $np_n \to \lambda$ as $n \to \infty$, then $X_n \xrightarrow{d} Y \sim \mathrm{Pois}(\lambda)$

• $X \sim \mathrm{NegBinom}(r,p)$ $\begin{cases} p \in [0,1] \\ r \in \mathbb{N}^+ \end{cases}$ X: # ind. Bern(p) trials to obtain r-th success; $p(x) = \binom{x-1}{r-1}p^r(1-p)^{x-r}$, $x = r, r+1, \dots$

$Y = X - r$; # failures before r-th success (X counts success/failures)

$p(y) = (-1)^y\binom{-r}{y}p^r(1-p)^y$, $y = 0, 1, \dots$; $EY = \frac{r(1-p)}{p}$, $VY = \frac{r(1-p)}{p^2}$

Prop. $\mu = EY$; $\mathrm{var}Y = \frac{1}{r}\mu^2 + \mu$; approx: $\begin{cases} r \to \infty, p \to 1 \\ r(1-p) \to \lambda \end{cases} \Rightarrow \begin{cases} EY = \lambda \\ VY = \lambda \end{cases} \Rightarrow Y \xrightarrow{d} \mathrm{Pois}(\lambda)$

• $X \sim \mathrm{Geo}(p) = \mathrm{NegBinom}(p, r=1)$: # trials before first success

$p(x) = p(1-p)^{x-1}$, $x = 1, 2, \dots$; $EX = (1-p)/p$; $VX = (1-p)/p^2$; **Prop.** if $s > t$, (memoryless) $P(X > s | X > t) = P(X > s - t)$

• $X \sim \mathrm{unif}([a,b])$, $a < b$: $f(x) = \frac{1}{b-a}$, $x \in [a,b]$ $\begin{cases} EX = (b+a)/2 \\ VX = (b-a)^2/12 \end{cases}$

• $X \sim \mathrm{Gamma}(\alpha, \beta)$ $\begin{cases} \alpha > 0: \text{shape} \\ \beta > 0: \text{scale} \end{cases}$ $\Gamma(\alpha) = \int_0^{\infty} t^{\alpha-1}e^{-t}\,dt$, $\alpha > 0$; $\Gamma(\alpha)\beta^{\alpha} = \int_0^{\infty} x^{\alpha-1}e^{-x/\beta}\,dx$

$f(x) = \frac{1}{\Gamma(\alpha)\beta^{\alpha}} x^{\alpha-1}e^{-x/\beta}$, $x > 0$; $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$

$EX = \alpha\beta$, $VX = \alpha\beta^2$; $\frac{X}{\beta} \sim \mathrm{Gam}(\alpha, 1)$; $\Gamma(n) = (n-1)\Gamma(n-1) = (n-1)!$

$M_X(t) = \left(\frac{1}{1-\beta t}\right)^{\alpha}$, $t < 1/\beta$

Prop. $\forall x > 0$ $P(X \le x) = P(Y \ge \alpha)$ for $X \sim \mathrm{Gam}(\alpha, \beta)$, $Y \sim \mathrm{Pois}(x/\beta)$

• $\chi^2_p = X \sim \mathrm{Gam}(p/2, 2)$; $f(x) = \frac{1}{\Gamma(p/2)2^{p/2}} x^{p/2 - 1}e^{-x/2}$ $\begin{cases} x > 0 \\ p \in \mathbb{Z} \end{cases}$

• $X \sim \mathrm{Exp}(\beta) = \mathrm{Gam}(1, \beta > 0)$; $f(x) = \frac{1}{\beta}e^{-x/\beta}$, $x > 0$ | maintains memoryless prop

Thm. $X \sim \mathrm{Exp}(\beta > 0)$, $Y = X^{1/\gamma}$, $\gamma > 0$. Then $Y \sim \mathrm{Weibull}(\gamma, \beta)$

$F_Y(y) = 1 - e^{-y^{\gamma}/\beta}$; $f(y) = \frac{\gamma}{\beta}y^{\gamma-1}e^{-y^{\gamma}/\beta}$, $0 < y < \infty$; $M_X(t) = e^{\mu t + \frac{\sigma^2 t^2}{2}}$

• $X \sim \mathrm{Normal}(\mu, \sigma^2)$ $\begin{cases} \mu \in \mathbb{R} \\ \sigma > 0 \end{cases}$ $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right)$, $x \in \mathbb{R}$

$\int f(x)\,dx = 1$, $\int x f(x)\,dx = \mu$, $\int x^2 f(x)\,dx = \mu^2 + \sigma^2$

$Z = \frac{X - \mu}{\sigma} \sim N(0,1)$; $f(z) = \frac{1}{\sqrt{2\pi}}\exp(-\frac{1}{2}z^2)$, $z \in \mathbb{R}$ $\begin{cases} EZ = 0 \\ VZ = 1 \end{cases}$

$\int_{\mathbb{R}} e^{-1/2 \cdot y^2}\,dy = \sqrt{2\pi} \Rightarrow \frac{1}{\sqrt{2}}\int_{\mathbb{R}} \exp(-\frac{1}{2}x^2)\,dx = \sqrt{\pi}$; $\Gamma(1/2) = \sqrt{\pi}$; $Y = Z^2 \sim \chi^2_1$

approx to Bin: $X \sim \mathrm{Bin}(n,p)$, $\begin{cases} np \to \infty \\ np(1-p) \to \infty \end{cases}$ then $Y \approx N(np, np(1-p))$

cont correction: $P(X = k) \approx P(k - 0.5 < Y < k + 0.5)$; $P(X \le k) \approx P(Y \le k + 0.5)$

• $X \sim \mathrm{Beta}(\alpha, \beta)$ $\begin{cases} \alpha > 0 \\ \beta > 0 \end{cases}$; $f(x) = \frac{1}{B(\alpha,\beta)}x^{\alpha-1}(1-x)^{\beta-1}$, $x \in (0,1)$

$EX^n = \frac{\Gamma(\alpha+\beta)\Gamma(\alpha+n)}{\Gamma(\alpha)\Gamma(\alpha+\beta+n)}$; $EX = \frac{\alpha}{\alpha+\beta}$, $VX = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$

$B(\alpha,\beta) = \int_0^1 x^{\alpha-1}(1-x)^{\beta-1} = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$; $X \sim \mathrm{Cauchy}(\theta)$, $\theta \in \mathbb{R}$

$f(x) = \frac{1}{\pi}\frac{1}{1 + (x-\theta)^2}$, $x \in \mathbb{R}$

$Y \sim N(\mu, \sigma^2)$, $X = e^Y \sim \mathrm{LogN}(\mu_Y, \sigma_Y)$ ⟺ $Y = \log X \sim N(\mu_Y, \sigma_Y)$

$f(x|\sigma_Y, \mu_Y) = \frac{1}{\sqrt{2\pi\sigma_Y^2}}\frac{1}{x}\exp\left(\frac{-1}{2\sigma_Y^2}(\log x - \mu_Y)^2\right)$

$EX = \infty$ (DNE), bell-shaped; **Prop.** $X, Y \overset{iid}{\sim} N(0,1) \Rightarrow \frac{X}{Y} \sim \mathrm{Cauchy}(0)$

Exp. Family: $f(x|\theta) = c(\theta)h(x)\exp\left\{\sum_{i=1}^{k} w_i(\theta)t_i(x)\right\} = c(\theta)h(x)\exp\langle w(\theta), t(x)\rangle$

$\int_{\mathbb{R}} f(x|\theta) = 1 \Rightarrow \frac{1}{c(w)} = \int_{\mathcal{X}} h(x)\exp\langle w(\theta), t(x)\rangle\,dx$; $\exp(x\ln(\frac{A}{B})) = (\frac{A}{B})^x$

$E\left[\sum_{i=1}^{k}\frac{\partial w_i(\theta)}{\partial \theta_j}t_i(X)\right] = \frac{\partial}{\partial\theta_j}\log\left(\frac{1}{c(\theta)}\right)$, $j = 1, \dots, k$ | differentiating log of normalizing const. wrt a parameter results in simple expectations of sufficient stats; $\log(a^k) = k\log(a)$, natural

closed under multiplication and reparametrization, not addition; mixtures arise addition at density level (no leading in algebra to analysis!)

Satisfies:
• finite-dimensional param.
• fixed support (not depending on unknown params)
• single exp form (no sums of exponentials)

3 (cont.)

ℓ-s family of dist. | μ: location param. σ: scale param.

$g(x|\mu,\sigma) = \frac{1}{\sigma}f(\frac{x-\mu}{\sigma})$, $x \in \mathbb{R}$ valid pdf

ℓ-fam: $\{g(x|\mu) = f(x-\mu) \mid \mu \in \mathbb{R}\}$

s-fam: $\{g(x|\sigma) = \frac{1}{\sigma}g(x/\sigma) \mid \sigma > 0\}$

Thm. $X \sim f \Rightarrow X + \mu \sim g(x) = f(x - \mu)$

$X \sim f \Rightarrow \sigma X \sim g(x) = \frac{1}{\sigma}f(x/\sigma)$

$X \sim f \Rightarrow \sigma X + \mu \sim g(x) = \frac{1}{\sigma}f(\frac{x-\mu}{\sigma})$

Thm. Let $Y \sim f$. Then $g(x|\mu,\sigma) \Rightarrow EX = \sigma EY + \mu$; $VX = \sigma^2 V(Y)$

$\lim_{r\to\infty}(1 + \frac{a}{r})^r = e^a$

$\lim_{r\to\infty}(1 + \frac{a_r}{r})^r = e^{\lim a_r}$ if $a_r \to a$

$e^a = \sum_{t=0}^{\infty}\frac{a^t}{t!}$

Markov: $P(g(X) \ge r) \le \frac{Eg(X)}{r}\ \forall r > 0$ ⟺ $Eg(X) \ge r \cdot P(g(X) \ge r)$; e.g. $P(|X - \mu| > 10) \le 1/4$

Chebyshev: X-RV, $g(x) \ge 0\ \forall x$;

$Y = g(X) = (X-\mu)^2$; $EY = \sigma^2$

$P(|X - \mu| \ge t) \le \sigma^2/t^2$ $(r = t^2)$; $P(|Y| \ge \sqrt{r}) \le \sigma^2/r$; $P(Y \ge r) \le EY/r$

Thm tighter norm. tail bnd.

$Z \sim N(0,1)$. For $t > 0$, $P(|Z| \ge t) \le \sqrt{\frac{2}{\pi}}\frac{1}{t}\exp(-\frac{t^2}{2})$

Prop. $\sqrt{\frac{2}{\pi}}\frac{e^{-t^2/2}}{t} \ll \frac{1}{t^2}$

Identities: • $X \sim \mathrm{Poisson}(\lambda)$: $P(X = 0) = \exp(-\lambda)$; $P(X = x+1) = e^{-\lambda}\frac{\lambda^{x+1}}{(x+1)!} = P(X = x)[\lambda/(x+1)]$

• $X \sim \mathrm{Gamma}(\alpha, \beta)$: $\begin{cases} f(x|\alpha,\beta) \\ \alpha > 1, \beta > 0 \end{cases}$ $\forall a, b$: $P(X_{\alpha,\beta} \in (a,b)) = \beta[f(a|\alpha,\beta) - f(b|\alpha,\beta)] + P(X_{\alpha-1,\beta} \in (a,b))$

• For $\alpha \in \mathbb{N}$, then recurse: $\mathrm{Gam}(\alpha-1,\beta) \to \mathrm{Gam}(\alpha-2,\beta) \to \dots \to \mathrm{Gam}(1,\beta) = \mathrm{Exp}(\beta)$

Steins: $X \sim N(\theta, \sigma^2)$, g: differentiable s.t. $E|g'(X)| < \infty \Rightarrow E[g(X)(X - \theta)] = \sigma^2 E[g'(X)]$

Hwang's identities: Let g: function, $|Eg(X)| < \infty$, $|g(-1)| < \infty$

• $X \sim \mathrm{Pois}(\lambda)$. Then $E[\lambda g(X)] = E[Xg(X-1)]$

• $X \sim \mathrm{NegBin}(r,p)$. $E[(1-p)g(X)] = E\left[\frac{X}{r+X-1}g(X-1)\right]$

Ch. 4: Multiple RV's

$S \xrightarrow{X} X(s) \in \mathbb{R}^n$, e.g. 2 (bivariate)

Discrete: $f(x,y) = P(X = x, Y = y) =: f_{XY}(x,y)$ (joint pmf/pdf)

$P((X,Y) \in A) = \sum_{(x,y) \in A} f(x,y)$; $E[g(X,Y)] = \sum_{(x,y) \in \mathbb{R}^2} g(x,y)f_{XY}(x,y)$

$E[ag_1(X,Y) + bg_2(X,Y)] = aE[g_1(X,Y)] + bE[g_2(X,Y)]$

marginal $f_X(x) = \sum_y f_{XY}(x,y)$; $f_Y(y) = \sum_x f_{XY}(x,y)$

* joint dist. determines marginal dist. but not vice versa

continuous: $P(X \in A) = \iint_{(x,y) \in A} f_{XY}(x,y)\,dx\,dy$

marginal: $f_X(x) = \int_{\mathbb{R}} f_{XY}(x,y)\,dy$; $f_Y(y) = \int_{\mathbb{R}} f_{XY}(x,y)\,dx$

$E[g(X,Y)] = \iint_{\mathbb{R}} g(x,y)f_{XY}(x,y)\,dx\,dy$

$F_{XY}(x,y) = P(X \le x, Y \le y) = \int_{-\infty}^{x}\int_{-\infty}^{y} f_{XY}(s,t)\,ds\,dt$

Domain

$\frac{\partial^2 F_{XY}(x,y)}{\partial x \partial y} = f_{XY}(x,y)$ @ cont. points (x,y) of $f_{XY}(x,y)$

Discrete: let $A = \{x\}$, B: $\forall y$: $P(Y \in B | X \in A) = \frac{P(X \in A \cap Y \in B)}{P(X \in A)} = \sum_{x \in A}\sum_{y \in B} f_{XY}(x,y) / \sum_{x \in A} f_X(x)$

$f(y|x) = P(Y = y | X = x) = \frac{f_{XY}(x,y)}{f_X(x)} = \frac{f_{XY}(x,y)}{\sum_y f_{XY}(x,y)}$; $\sum_y f(y|x) = 1$

Prop. $f_{XY}(x,y) = f_X(x)f_{Y|X}(y|x) = f_Y(y)f_{X|Y}(x|y)$ whenever $f(x), f(y) > 0$

cont.:

$P(Y \in B | X = x) = \lim_{\epsilon \downarrow 0} P(Y \in B | X \in (x - \epsilon, x + \epsilon)) = \int_B \frac{f_{XY}(x,y)}{f_X(x)}\,dy$

$f_{Y|X}(y|x) = f_{XY}(x,y)/f_X(x) = f_{XY}(x,y)/\int_y f_{XY}(x,y)\,dy$

$E[g(Y)|X] = \int g(y)f_{Y|X}(y|x)\,dy$; often easier to identify joint dist. via (*)

Def: X and Y ind. if $f_{XY}(x,y) = f_X(x)f_Y(y)\ \forall x, y \in \mathbb{R} \Rightarrow f(y|x) = f(y)$

• identify (x,y) or (A,B) that violate ↑ for non-independence $\Rightarrow P(Y \in B | X \in A) = P(Y \in B)$

Prop. $X \perp\!\!\!\perp Y$ i.f.f. $f_{XY}(x,y) = g(x)h(y)\ \forall x, y \in \mathbb{R}$

Thm. $E[g(X)h(Y)] = E g(X) E g(Y)$; $X \perp\!\!\!\perp Y$, $Z = X + Y$. $M_Z(t) = M_X(t)M_Y(t)$ $\forall t$

Prop. $X \sim N(\mu_1, \sigma_1^2)$, $Y \sim N(\mu_2, \sigma_2^2)$, $X \perp\!\!\!\perp Y$; $Z = X + Y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$

Markov: $P(g(X) \ge r) \le \frac{E[g(X)]}{r}$

$g(X)$: $(X - E(X))^2 \to P(|X - E(X)| \ge t) \le \frac{\mathrm{var}X}{t^2}$

$g(X) = \frac{(X-\mu)^2}{\sigma} \to P(|X - \mu| \ge \sigma\sqrt{r}) \le 1/r$

$\Leftrightarrow P(|X - \mu| \ge t\sigma) \le 1/t^2$, letting $r = t^2$

Ch. 4 (cont)

Prop $\forall A \subset \mathbb{R}^2$, $P((U,V)\in A) = P((X,Y)\in g^{-1}(A))$

$g^{-1}(A) = \{(x,y) \mid g(x,y)\in A\}$; Discrete: $f_{U,V}(u,v) = \sum f_{X,Y}(x,y)$ over x,y: $g_1(x,y)=u$, $g_2(x,y)=v$

cont. $g:\mathbb{R}^2\to\mathbb{R}^2$ 1-1 mapping: $g^{-1}(\{u,v\})$ has at most one element

$g^{-1}(\{u,v\}) = h(u,v) = \binom{h_1(u,v)}{h_2(u,v)}$

$f_{U,V}(u,v) = f_{X,Y}(h_1(u,v), h_2(u,v))\,|J|$

$J = \left|\frac{\partial(x,y)}{\partial(u,v)}\right| = \begin{vmatrix} \partial x/\partial u & \partial x/\partial v \\ \partial y/\partial u & \partial y/\partial v\end{vmatrix}$, $|J| = ad - bc$

$(X,Y)\to g(X,Y)\in\mathbb{R}^2$, many-to-one; partition $A = \{(x,y) \mid f_{X,Y}(x,y)>0\} = A_1\cup\dots\cup A_k$ s.t. $A_i \to g(A_i)$ 1-1, $i=1,\dots,k$. Then $f_{U,V}(u,v) = \sum_{i=1}^{k} f_{XY}(h_{1i}(u,v), h_{2i}(u,v))|J_i|$

Thm. X,Y - RV, then $EX = E[E(X|Y)]$ | iterated expectation

Mixture Dist: (of X) depends on RV Y, e.g.

$X|Y \sim \text{Binomial}(Y,p)$, $Y\sim\text{Pois}(\lambda)$ | mixture of Bin dist's; Pois mixing mechanism

$P(X=x) = \sum_{y\in\mathbb{N}} P(X=x, Y=y) = \sum_{y\in\mathbb{N}} P(Y=y)P(X=x|Y=y) = \sum \frac{e^{-\lambda}\lambda^y}{y!}\binom{y}{x}p^x(1-p)^{y-x}$, $x\in\mathbb{N}$

Prop. $Y\in\{1,\dots,k\}$, $P(Y=i)=p_i$, $i=1,\dots,k$, $X|Y=i \sim N(\mu_i,\sigma_i^2)$

$f_X(x) = \sum p_i N(x|\mu_i,\sigma_i^2) = \sum p_i \frac{1}{\sqrt{2\pi}\sigma_i}\exp\left(\frac{-1}{2\sigma_i^2}(x-\mu_i)^2\right)$

(mixture of k normal components)

Hierarchical: X dist. obtained w/ many stages of cond. dist's

e.g. $X|Y\sim\text{Bin}(Y,p)$, $Y|\Lambda\sim\text{Pois}(\Lambda)$, $\Lambda=\text{EXP}(\beta)$

$(\Rightarrow) Y|\Lambda = \lambda \sim \text{Pois}(\lambda)$

$P(Y=y) = \int_0^\infty f_{Y,\Lambda}(y,\lambda)\,d\lambda = \int_0^\infty P(Y=y|\Lambda=\lambda) f_\Lambda(\lambda)\,d\lambda$ (marginalize)

$= \int_0^\infty e^{-\lambda}\frac{\lambda^y}{y!}\frac{1}{\beta}e^{-\lambda/\beta}\,d\lambda$

$\Rightarrow$ | $X|Y\sim\text{Bin}(Y,p)$, $Y\sim\text{NegBin}(\frac{1}{1+\beta},1)$

$\sim \text{NegBin}(\frac{1}{1+\beta},1)$

$X|\theta\sim\text{Bin}(n,\theta)$, $\theta\in(0,1)$

$\theta\sim\text{Beta}(\alpha,\beta)$

$P(X=x) = \int P(X=x|\theta=\pi) f_\theta(\pi)\,d\pi$ (marginalize)

$= \int \binom{n}{x}\theta^x(1-\theta)^{n-x}\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\theta^{\alpha-1}(1-\theta)^{\beta-1}\,d\theta$

$\text{Var}X = E[\text{Var}[X|Y]] + \text{Var}[E[X|Y]]$

$P(A) = E(1_A)$ ∀ event A where $1_A = \begin{cases}1, & A \text{ occurs}\\0, & \text{otherwise}\end{cases}$

$\phi(t) = \frac{1}{\sqrt{2\pi}}\exp\left(\frac{-t^2}{2}\right)$; $X\sim N(\mu,\sigma^2) \Rightarrow f(x) = \phi\left(\frac{x-\mu}{\sigma}\right)$

$\text{cov}(X,Y) = E[(X-EX)(Y-EY)]$; $\text{corr}(X,Y) = \frac{\text{cov}(X,Y)}{\sqrt{\text{Var}X}\sqrt{\text{Var}Y}}$ standardized covariance

$\text{cov}(X,X) = \text{Var}(X)$; $EX=EY=0 \Rightarrow \text{cov}(X,Y) = E(XY)$; $\text{cov}(X,cX) = c\,\text{Var}(X)$

CSI: $E[XY] \le \sqrt{E(X^2)}\sqrt{E(Y^2)} \Leftrightarrow [E(XY)]^2 \le E(X^2)E(Y^2)$

Equality i.f.f. $Y \overset{as}{=} cX \Leftrightarrow P(Y=cX)=1$, c const $\in\mathbb{R}$

Prop. Let $U = X-EX$, $V = Y-EY$

$[E(UV)]^2 \le E(U^2)E(V^2) \Leftrightarrow |\text{corr}(X,Y)|\le 1$

Equality i.f.f. $U \overset{as}{=} cV \Rightarrow P(U=cV)=1 \Rightarrow |\text{corr}(X,Y)|=1$

Thm. $\text{cov}(X,Y) = E(XY) - E(X)E(Y)$;

$X\perp\!\!\!\perp Y \Rightarrow \text{cov}(X,Y) = \text{corr}(X,Y) = 0$ ($\not\Leftarrow$ not necessarily)

Thm. $\forall a,b\in\mathbb{R}$, $\text{cov}(X,A+B) = \text{cov}(X,A) + \text{cov}(X,B)$ (linearity)

$V(aX+bY) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab\,\text{cov}(X,Y)$

Thm. $\text{corr}(X,Y)=1 \Leftrightarrow Y=bX+c$, $b>0$, $c\in\mathbb{R}$

$\text{corr}(X,Y)=-1 \Leftrightarrow Y=bX+c$, $b<0$, $c\in\mathbb{R}$

Bivariate Normal: $Z=(X,Y)\sim N(\mu,\Sigma)$ | $\mu\in\mathbb{R}^2$, $\Sigma\in\mathbb{R}^{2\times2}$

$f(z) = f(x,y) = \frac{1}{(2\pi)^{2/2}|\Sigma|^{1/2}}\exp\left\{-\frac{1}{2}(z-\mu)^T\Sigma^{-1}(z-\mu)\right\}$

$E(Z) = E\left[\binom{X}{Y}\right] = \binom{\mu_1}{\mu_2}$; $\text{cov}Z = \Sigma = \begin{pmatrix}\Sigma_{11} & \Sigma_{12}\\ \Sigma_{21} & \Sigma_{22}\end{pmatrix}$ | $\Sigma_{11}=VX$, $\Sigma_{22}=VY$, $\Sigma_{12}=\Sigma_{21}=\text{cov}(X,Y)$

$= E[(Z-EZ)(Z-EZ)^T]$

notation: $\sigma_{XY} = \text{cov}(X,Y)$, $\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X\sigma_Y}$ ($\text{corr}(X,Y)$)

marginal: $X\sim N(\mu_1,\Sigma_{11})$, $Y\sim N(\mu_2,\Sigma_{22})$

Identities: $\iint f(x,y)\,dx\,dy = \mu_1$

- $\iint f(x,y)\,dx\,dy = 1$
- $\iint y f(x,y)\,dx\,dy = \mu_2$
- $\iint (z-\mu)(z-\mu)^T f(z)\,dz = \Sigma$

Thm. $\text{cov}(X,Y)=0 \Rightarrow X\perp\!\!\!\perp Y$

- $Y|X=x \sim N[E(Y|X), \text{Var}(Y|X)]$

$E(Y|X) = \mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(x-\mu_1)$; $V(Y|X) = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}$

$\int_{\mathbb{R}} e^{-|x|}\,dx = 2$; symmetric abt a: $f(a+x) = f(a-x)$ $\forall x>0$

$\int e^{-x^2}dx = \sqrt{\pi}$; $\int e^{-\frac{1}{2}z^2}dz = \sqrt{2\pi}$, $x = z/\sqrt{2}$

$\int e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}dx = \sqrt{2\pi}\sigma$; $z = \frac{x}{\sigma}$

$f(x) = f(x_1,\dots,x_n) = P(X_1=x_1,\dots,X_n=x_n)$

$P(X\in A) = \sum_{x\in A}P(X=x) = \sum_{x\in A} f(x)$

$P(X\in A) = \int_A f(x)\,dx = \int\dots\int f(x_1,\dots,x_n)\,dx_1\dots dx_n$ over $(x_1,\dots,x_n)\in A$

$E g(X) = \begin{cases}\int_{\mathbb{R}^n} g(x)f(x)\,dx\\ \sum_{x\in\mathbb{R}^n} g(x)f(x)\end{cases}$

marginal: $f_{X_1,\dots,X_k}(x_1,\dots,x_k) = \begin{cases}\int_{\mathbb{R}}\dots\int_{\mathbb{R}} f(x_1,\dots,x_n)\,dx_{k+1}\dots dx_n\\ \sum_{x_{k+1}}\dots\sum_{x_n} f_{X_1,\dots,X_n}(x_1,\dots,x_n)\end{cases}$

e.g. $f(x_1,x_2) = \int_{\mathbb{R}}\int_{\mathbb{R}} f(x_1,\dots,x_4)\,dx_3\,dx_4$

Conditional: $f(x_{k+1},\dots,x_n|x_1,\dots,x_k) = \frac{f(x_1,\dots,x_k,x_{k+1},\dots,x_n)}{f(x_1,\dots,x_k)}$

$\to$ admits factorization; joint = marginal x cond

Chain: $f(x_1,\dots,x_n) = f(x_1)f(x_2|x_1)\cdots f(x_n|x_1,\dots,x_{n-1})$

mutual ind. $f(x_1,\dots,x_n) = \prod_{i=1}^{n} f_{X_i}(x_i)$

$E[g_1(X_1)\cdots g_n(X_n)] = E[g_1(X_1)]\cdots E[g_n(X_n)]$

Prop. X, Y discrete, Z, T cont

$f(x,y,z,t) = f(x,y)f(z,t|x,y) = f(z,t)f(x,y|z,t)$ (pmf, cond. pdf, pdf, cond. pmf)

$P(X\in A, Y\in B, Z\in C, T\in D) = \sum_{x\in A}\sum_{y\in B}\int_D\int_C f(x,y,z,t)\,dz\,dt$

Multinomial (trials; pmf; multinomial coefficient) | m ind. trials, n possible outcomes w/ cell prob's $(p_1,\dots,p_n)$, $\sum_{i=1}^n p_i = 1$ | $(X_1,\dots,X_n)$: counts of each of n outcomes s.t. $x_1+\dots+x_n = m$

$f(x_1,\dots,x_n) = \binom{m}{x_1,\dots,x_n}p_1^{x_1}\cdots p_n^{x_n} = \frac{m!}{x_1!\cdots x_n!}p_1^{x_1}\cdots p_n^{x_n}$ | $x_1,\dots,x_n\in\mathbb{N}^n$, $\sum_{i=1}^n x_i = m$

$\sum_{x_1+\dots+x_n=m}\binom{m}{x_1,\dots,x_n}p_1^{x_1}\cdots p_n^{x_n} = (p_1+\dots+p_n)^m = 1$ multinomial formula

marginal: $f(x_n) = \sum_{x_1+\dots+x_{n-1}=m-x_n} f(x_1,\dots,x_n) = \frac{m!}{x_n!(m-x_n)!}p_n^{x_n}(1-p_n)^{m-x_n} \sim \text{Bin}(m,p_n)$

cond: $f(x_1,\dots,x_{n-1}|x_n) = \frac{f(x_1,\dots,x_n)}{f(x_n)} = \frac{(m-x_n)!}{x_1!\cdots x_{n-1}!}\left(\frac{p_1}{1-p_n}\right)^{x_1}\cdots\left(\frac{p_{n-1}}{1-p_n}\right)^{x_{n-1}} \sim \text{Mult}\left(m-x_n \text{ trials}, \left(\frac{p_1}{1-p_n},\dots,\frac{p_{n-1}}{1-p_n}\right)\right)$

invariance prop (marg/cond.) holds for many exponential fam dist (e.g. normal)

Thm. $X_1,\dots,X_n$ mutually ind., $Z = X_1+\dots+X_n$; $M_Z(t) = M_{X_1}(t)\cdots M_{X_n}(t)$; | if $X_i \overset{iid}{\sim}$, $i=1,\dots,n$ then $M_Z(t) = [M_{X_1}(t)]^n$

Prop. $X_1,\dots,X_n$ mutually ind. $\Leftrightarrow f(x_1,\dots,x_n) = g_1(x_1)\cdots g_n(x_n)$ for some functions $g_1,\dots,g_n$

$\Rightarrow g_1,\dots,g_n$ are also mutually ind

Prop. $X_i\sim\text{Gam}(\alpha_i,\beta)$, $X_i \overset{ind}{\sim}$, $i=1,\dots,n$. Then $X_1+\dots+X_n\sim\text{Gam}(\alpha_1+\dots+\alpha_n,\beta)$

Prop. $X_i \overset{ind}{\sim} N(\mu_i,\sigma_i^2)$ | $i\in\{1,\dots,n\}$, $a_i, b_i\in\mathbb{R}$. Then $Z = \sum_{i=1}^n(a_iX_i+b_i) \sim N\left(\sum_{i=1}^n(a_i\mu_i+b_i), \sum_{i=1}^n a_i^2\sigma_i^2\right)$

$\sum_{n=1}^\infty \frac{1}{n} = \infty$ (Harmonic series diverge)

Ch. 5: Random Sample Properties

$X_1,\dots,X_n \overset{iid}{\sim} f$. Then f: (n-iid) sample

Statistic: $Y = T(X_1,\dots,X_n)$

e.g. $\bar{X} = \frac{1}{n}(X_1+\dots+X_n)$, $S^2 = \frac{1}{n-1}\sum(X_i-\bar{X})^2$

$X_{(k)} = \min\{X_1,\dots,X_n\}\setminus\{X_{(1)},\dots,X_{(k-1)}\}$

k-th smallest number; order stat: $(X_{(1)},\dots,X_{(n)})$

Thm. Let $(X_1,\dots,X_n) \overset{iid}{\sim}$ with $\mu,\sigma^2<\infty$;

Then $E\bar{X}=\mu$, $\text{Var}\bar{X} = \sigma^2/n$, $ES^2 = \sigma^2$

- $\bar{X}$ and S^2 unbiased estimates

Dist. of $\bar{X}$: (1) moments, (2) convolution

(1) $M_{\bar{X}}(t) = Ee^{t\bar{X}} = (M_X(t/n))^n$; easy if we can recognize $\bar{X}$ from known family of dist

(2) $X\sim f_X$, $Y\sim f_Y$, $Z = X+Y$, $X\perp\!\!\!\perp Y$

$f_Z(z) = \int_{\mathbb{R}} f_X(x)f_Y(z-x)\,dx$; useful when $f(x_1,\dots,x_n)$ does not belong to known/well-recognized family

$f(z) = \sum_x f_X(x)f_Y(z-x)$

(2) $Z = X+Y$: $(X,Y)\to(X+Y,X)$

$Z = X-Y$: $(X,Y)\to(X-Y,X)$ and $\int_{\mathbb{R}} f_X(x)f_Y(x-z)\,dx = f(z)$

- $Z = XY$: $(X,Y)\to(XY,X)$, etc.

$\sum_{i=1}^n(X_i-\bar{X}) = \sum X_i - n\bar{X} = \sum X_i - \sum X_i = 0$

Thm. Let $X_i \overset{iid}{\sim} N(\mu,\sigma^2)$, $i=1,\dots,n$

- $\bar{X}\perp\!\!\!\perp S^2$; $\bar{X}\sim N(\mu,\sigma^2/n)$; $E(X_i - E(\mu)) = 0$
- $(n-1)\frac{S^2}{\sigma^2}\sim\chi^2_{n-1}$; $\text{Var}(X\sim\chi^2_k) = 2k$

$T := (\bar{X}-\mu)/(S/\sqrt{n})$, $t\in\mathbb{R}$

$f(t) = \frac{\Gamma[(p+1)/2]}{\Gamma(p/2)}\frac{1}{(p\pi)^{1/2}}\frac{1}{(1+\frac{t^2}{p})^{(p+1)/2}}$

- T: ratio of 2 ind. RVs: $\bar{X}-\mu\sim N$, $\sqrt{(n-1)S^2}\sim\sqrt{\chi^2_{n-1}}$

$n=2 \Rightarrow T = \frac{Y-m}{|Y-m|}$ ... (bounded)

$X_n \xrightarrow{d} X := F_n(x)\to F_X(x)$ at all cont. points for F_X | $\Phi(x) = P(Z\le x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}}e^{-t^2/2}dt$

$X_n \xrightarrow{P} X := \forall\varepsilon>0$, $P(|X_n-X|\ge\varepsilon) \Leftrightarrow P(s: |X_n(s)-X(s)|\ge\varepsilon)\to0$ as $n\to\infty$

Thm (WLLN): $X_1,\dots,X_n \overset{iid}{\sim}$ with $\begin{vmatrix}EX_i=\mu\\ VX_i=\sigma^2<\infty\end{vmatrix}$. Then $\bar{X}_n\xrightarrow{P}\mu$; SLLN: $\bar{X}_n\xrightarrow{as}\mu$.

$X_n\xrightarrow{a.s.}X := P(\lim_{n\to\infty}|X_n-X|=0) = 1 \Leftrightarrow P(\lim_{n\to\infty}X_n = X) = 1 \Leftrightarrow \forall\varepsilon>0$, $P(\lim_{n\to\infty}|X_n-X|<\varepsilon) = 1$

Prop. $X_n\xrightarrow{as}X \Rightarrow X_n\xrightarrow{P}X$; $X_n\xrightarrow{P}X\Rightarrow X_n\xrightarrow{d}X$; Fatou's L: $E[\liminf_{n\to\infty}X_n]\le\liminf_{n\to\infty}E(X_n)$

Thm (CLT): $X_i \overset{iid}{\sim}$, $i=1,\dots,n$ (letting $\begin{vmatrix}\mu = EX_i, Z\sim N(0,1)\\ \sigma^2 = VX_i > 0\end{vmatrix}$ ($\Rightarrow VX_i<\infty$ (mild condition))). Then $\left(\frac{\bar{X}_n-\mu}{\sigma/\sqrt{n}}\right) = \sqrt{n}\left(\frac{\bar{X}_n-\mu}{\sigma}\right)\xrightarrow{d}N(0,1)$.

Cor. Let $Y_i = (X_i-\mu)/\sigma$ s.t. $\begin{vmatrix}EY_i=0\\ EY_i^2=1\end{vmatrix}$. Then $\frac{1}{\sqrt{n}}(Y_1+\dots+Y_n)\xrightarrow{d}N(0,1)$

Prop. $X_n\xrightarrow{as}X$, $Y_n\xrightarrow{as}b\in\mathbb{R}\Rightarrow aX_n+Y_n\xrightarrow{as}aX+b$

$X_n\xrightarrow{P}X$, $Y_n\xrightarrow{P}b\in\mathbb{R}\Rightarrow aX_n+Y_n\xrightarrow{P}aX+b$; $X_n\xrightarrow{d}X$, $Y_n\xrightarrow{P}b\in\mathbb{R}\Rightarrow aX_n+Y_n\xrightarrow{d}aX+b$

Final

3) Exp family: $f(x|\theta) = c(\theta)h(x)\exp(\sum w_i(\theta)t_i(x)) = c(\theta)h(x)\exp\langle w(\theta), t(x)\rangle$

NegBin(r,p); $P(x|p) = f(x|\theta)$ whereas $p(x|r,p)\ne f(x|\theta)$; cannot factorize $(x-r)!$ (separate)

$f(x|k,p,\alpha,\beta) = \sum p_i\,\text{Gam}(x|\alpha_i,\beta_i)\ne f(x|\theta)$; cannot rewrite $\sum_i p_i e^{a(y)}$ | Mixtures invalidate addition of density funcs

$P(X=x) = p_k \begin{vmatrix}k\in\mathbb{Z}\\ p_k\ge 1\end{vmatrix}$, co-ly many $\{p_k\}$; $f(x|\theta)$ requires finite-dim θ. P, Z, X - RV's (Hierarchy)

3) $X|_{P,Z=1}\sim N(\mu_1,1)$, $X|_{P,Z=0}\sim N(\mu_0,1)$, $Z|P\sim\text{Bernoulli}(P)$, $P\sim\text{Beta}(\alpha,\beta)$

$P(A) = E(1_A) \Rightarrow P(Z=1) = E[P(Z=1|P)] \Rightarrow Z\sim\text{Bern}(\alpha/(\alpha+\beta))$; $f(x|\theta) = \phi([x-\mu_i]/\sigma_i)$, $i\in\{0,1\}$.

$EX = E[E[X|Z]] = \mu_0P(Z=0) + \mu_1P(Z=1) = \mu_0[\beta/(\alpha+\beta)] + \mu_1[\alpha/(\alpha+\beta)]$; $Z|X\sim\text{Bern}(P(Z=1|X=x))$

$P(Z|X) = [f(x|z)f(z)]/f(x)|_{z=1} = [\alpha\phi(x-\mu_1)]/[\beta\phi(x-\mu_0)+\alpha\phi(x-\mu_1)]$ | any linear transformation of a jointly normal vector is jointly normal

1) Roll dice ind. n times; $X_i\in\{1,\dots,6\}$, $Y_i=0\Rightarrow X_i\in\{5,6\}$, $Y_i=1\Rightarrow X_i\in\{1,4\}$, $Y_i=-1\Rightarrow X_i\in\{2,3\}$, $i\in\mathbb{N}^n$

$S = Y_1+Y_2$; $P(S) = 1/9\,(S=2,-2)$, $2/9\,(S=1,-1)$, $1/3\,(S=0)$ | $P(Y_1Y_2) = P(Y_1)P(Y_2)$, $1/9(P(1))$, $1/9(P(-1))$

$S_n = \sum_{i=1}^n Y_i$; $E(S_n) = \sum_i E(Y_i) = 0$; $V(S_n) = n\cdot V(Y_1) = 2n/3$; CLT: $[S_n - ES_n]/\sqrt{V S_n} = S_n/\sqrt{2n/3}\xrightarrow{d}N(0,1)$ | $P(S_n\le t) \approx \Phi(t/\sqrt{2n/3})$

For $\varepsilon>0$, show $P(|Y_1\cdots Y_n|\ge\varepsilon)\le\varepsilon_n\to0$ as $n\to\infty$;

$|Y_1\cdots Y_n| \le |Y_1\cdots Y_n| = 1$ whenever $Y_i\ne0$ ∀ i

$T_i := 1\{Y_i\ne0\}$ s.t. $ET_i = 2/3$, $VT_i = 1/3(2/3) = 2/9$

$Z_n = \prod T_i$; (ε 0/1) $\Rightarrow EZ_n = 2/3$, $VZ_n = 2n/9$

$P(|Y_1\cdots Y_n|\ge\varepsilon)\le P(Z_n = 0)$ & bound

$Z_n = 0 \Rightarrow |Z_n - EZ_n| \ge EZ_n = 2/3$

$P(|Z_n - EZ_n|\ge 2/3)\le\frac{\text{Var}Z_n}{(2/3)^2} = \frac{2}{n}$

$\Rightarrow P(|Y_1\cdots Y_n|\ge\varepsilon)\le\frac{2}{n}\to0$ as $n\to\infty$

4) $X\sim N(\mu,\sigma^2)$, $\varepsilon = aX+Z$, $Z\sim N(0,1)\perp\!\!\!\perp X$, observed $Y = X+\varepsilon$

$\binom{X}{Y} = \begin{pmatrix}1 & 0\\ 1+a & 1\end{pmatrix}\binom{X}{Z} \Rightarrow \binom{X}{Y}$ is bivariate normal

$E\left[\binom{X}{Y}\right] = \binom{EX}{EY} = \binom{\mu}{(1+a)\mu}$; $\Sigma = \begin{pmatrix}VX & \text{cov}XY\\ \text{cov}XY & VY\end{pmatrix} = \begin{pmatrix}\sigma^2 & (1+a)\sigma^2\\ (1+a)\sigma^2 & (1+a)^2\sigma^2+1\end{pmatrix}$

$VY = V[(1+a)X+Z]$; $\text{cov}(X,Y) = \text{cov}(X,(1+a)X) + \text{cov}(X,Z) = (1+a)\sigma^2$ ... $(1+a)^2\sigma^2+1$

$\text{corr}(X,Y) = \frac{\sigma_{XY}}{\sigma_X\sigma_Y}$ ... $(1+a)\sigma^2 \Rightarrow (1+a)<0 \Leftrightarrow a<-1$

$E(Y|X) = \mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(x-\mu_1) = \mu_2 + \frac{\sigma_{XY}}{\sigma_X^2}(x-\mu_1)$

5) $U,V\sim\text{Uni}(0,1)$, $U\perp\!\!\!\perp V$, $f(u) = 1$, $f(v) = \begin{cases}1 & 0<u<1\\ 0 & 0<v<1\end{cases}$

$Y = U^2$; $F_Y(y) = P(U^2\le y) = P(U\le\sqrt{y}) = \sqrt{y}$

$f(y) = \frac{d}{dy}F(y) = 1/2\sqrt{y}$, $0<y<1$

$X = U+V$; $f(x) = \int_{\mathbb{R}} f(u)f_V(x-u)\,du = \int_{\mathbb{R}} f(u)1\{0<u<1\}f_V(x-u)1\{0<x-u<1\} = \int_{\max(0,x-1)\text{ - changes @ }x=1}^{\min(1,x)\text{ - changes @ }x=1} du$

$0<x<1$: $f(x) = \int_0^x du = x$; $1<x<2$: $f(x) = \int_{x-1}^1 du = (2-x)$

$Y=y\Rightarrow U=\sqrt{y}$ ($V\ge0$) $\Rightarrow X = \sqrt{y}+V\in[\sqrt{y},\sqrt{y}+1]$: $\{(x,y) \mid 0<y<1, \sqrt{y}<x<\sqrt{y}+1\}$

Choose $y = 1/4$, $x = 0.1$. Then $x = 0.1<\sqrt{1/4} = \frac{1}{2}\Rightarrow P(X=x, Y=y)$ undefined ($\to0$) since $x<\frac{1}{2}$, ... But $f_X(0.1)>0$, $f_Y(1/4)>0$ (product density on support) when $y = 1/4$,

and $f(x,y)\ne f(x)f(y)\Rightarrow X\not\perp\!\!\!\perp Y$

$\begin{vmatrix}X = U+V\\ Y = U^2\end{vmatrix}\to\begin{vmatrix}U = \sqrt{Y}\\ V = X-\sqrt{Y}\end{vmatrix}$, $J = \begin{pmatrix}0 & 1/2\sqrt{y}\\ 1 & -1/2\sqrt{y}\end{pmatrix}$, $|J| = 1/2\sqrt{y}$

$f(x,y) = f_U(\sqrt{y})f_V(x-\sqrt{y})\frac{1}{2\sqrt{y}} = \frac{1}{2\sqrt{y}}$ for $\begin{vmatrix}0<u<1\Rightarrow 0<\sqrt{y}<1\Rightarrow 0<y<1\\ 0<v<1\Rightarrow 0<x-\sqrt{y}<1\Rightarrow\sqrt{y}<x<\sqrt{y}+1\end{vmatrix}$

Review (1) | RV's and vectors | Mult. Normal Dist's

cdf: $:=P(X \le x) =: F(x)$ · $F(-\infty)=0$,
· nondecreasing, right cont., $F(\infty)=1$

pdf: $p(x)=P(X=x)$

$P(X\in B)=\sum_{x\in B} p(x) = \int_B f(x)\,dx$

· $f(x)=\frac{d}{dx}F(x)$; $F(x)=\int_0^x f(t)\,dt$

Discrete: unif, Bernoulli, Binomial, Hypergeom, Geom, Neg Bin, Poisson

Cont: unif, Exp, Gamma, chi-square, Normal (Gaussian), Lognormal, Beta, Weibull

(univariate) transformation: $Y=g(X)$

$$f_Y(y)=f_X(g^{-1}(y))\left|\frac{d}{dy}g^{-1}(y)\right|$$

Exp Fam: · finite-dim parameter · fixed support · single exp form

$f(x|\theta)=h(x)c(\theta)\exp\left(\sum_{i=1}^{k} w_i(\theta)t_i(x)\right)$

LS-Fam: $f(x|\mu,\sigma)=\frac{1}{\sigma}f\left(\frac{x-\mu}{\sigma}\right)$

LF: $f_\mu(x)=f_0(x-\mu)$; SF: $f_\sigma(x)=\frac{1}{\sigma}f_0\left(\frac{x}{\sigma}\right)$

Multivariate: joint cdf, pdf, pmf, etc.

multinomial: $P(X_1,\dots,X_k)=\frac{n!}{x_1!\cdots x_k!}p_1^{x_1}\cdots p_k^{x_k}$

$\sum_{i=1}^{p} x_i = n$, x_i: # trials that result in category i for n iid trials

$E(X_i)=np_i$, $\mathrm{cov}(X_i,X_j)=\begin{cases} np_i(1-p_i) & i=j \\ -np_ip_j & i\ne j\end{cases}$

Covariance: $\mathrm{cov}(X)=\mathrm{var}(X)=E[(X-\mu)(X-\mu)^T]$

· $\mathrm{cov}(X)=\mathrm{cov}(X)^T$, $\mathrm{cov}(a+X)=\mathrm{cov}(X)$

· $\mathrm{cov}(AX)=A\,\mathrm{cov}(X)A^T$, $a^T\mathrm{cov}(X)a\ge 0$

$\mathrm{cov}(X,Y)=E[(X-\mu_X)(Y-\mu_Y)^T]$

· $\mathrm{cov}(Y,X)=\mathrm{cov}(X,Y)^T$, $\mathrm{cov}(X,X)=\mathrm{cov}(X)$

· $\mathrm{cov}(AX,BY)=A\,\mathrm{cov}(X,Y)B^T$

$Z=\binom{X}{Y}$. $\mathrm{cov}(Z)=\begin{pmatrix}\mathrm{cov}(X) & \mathrm{cov}(X,Y)\\ \mathrm{cov}(Y,X) & \mathrm{cov}(Y)\end{pmatrix}$

· marginalize by summing/integrating f over remaining variables

Conditional: $f(y|x)=f(x,y)/f(x)$

Iterated mean/variance: $E(Y)=E[E[Y|X]]$,

$\mathrm{Var}(Y)=\mathrm{var}[E[Y|X]]+E[\mathrm{Var}[Y|X]]$

(multivariate) transformation: $X=(X_1,\dots,X_n)^T$,

$Y=g(X)$: 1-1 (invertible) trans. s.t. $Y=(Y_1,\dots,Y_n)^T$

$f_Y(y)=f_X(g^{-1}(y))\,|\det(J)|$ | $x=g^{-1}(y)$: inverse trans. J: Jacobian of x

Multivariate Normal (Recall): $M(t)=E(e^{tX})$

$N\sim(\mu,\sigma^2)\Rightarrow M(t)=\exp\left(t\mu+\frac{t^2\sigma^2}{2}\right)$

$N\sim(\mu,\Sigma)\Rightarrow M(t)=\exp\left(t^T\mu+\frac{t^T\Sigma t}{2}\right)$

Prop. $X\sim N(\mu,\Sigma)$ iff any linear combo $a^TX \sim N(a^T\mu, a^T\Sigma a)$

$X\sim N(\mu,\Sigma)\in\mathbb{R}^p$, Σ: full rank: $f(x)=\frac{1}{(2\pi)^{p/2}|\Sigma|^{1/2}}\exp\left\{-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right\}I(x\in\mathbb{R}^p)$

$\begin{pmatrix}a&b\\c&d\end{pmatrix}^{-1}=(ad-bc)^{-1}\begin{bmatrix}d&-b\\-c&a\end{bmatrix}$, $\Sigma=\begin{pmatrix}\sigma_X^2 & \sigma_{XY}\\ \sigma_{XY} & \sigma_Y^2\end{pmatrix}$, $\sigma_{XY}=\rho\sigma_X\sigma_Y$, ρ: $\mathrm{corr}(X,Y)$, $\Sigma^{-1}=\frac{1}{\sigma_X^2\sigma_Y^2-\rho\sigma_X^2\sigma_Y^2}\begin{pmatrix}\sigma_Y^2 & -\rho\sigma_X\sigma_Y\\ -\rho\sigma_X\sigma_Y & \sigma_X^2\end{pmatrix}$

Let $p=2$: $f(x,y)=\left[2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}\right]^{-1}\exp\left\{-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_X)^2}{\sigma_X^2}-\frac{2\rho(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y}+\frac{(y-\mu_Y)^2}{\sigma_Y^2}\right]\right\}$, $(x,y)\in\mathbb{R}^2$

Review (2): Random samples, Sampling Dist, convergence concepts, Delta Method

Random samples: Given n iid RV's, i.e. $X_1,\dots,X_n$, joint dist: $f(x_1,\dots,x_n)=\prod_{i=1}^n f(x_i)$ — and nothing else

Prop. Let $X_1,\dots,X_n$: random sample size n; let $Y=T(X_1,\dots,X_n)$:= statistic (function of data); sampling dist. (of Y) — Prob dist. of Y is

Statistics: $\bar{X}=\frac{1}{n}\sum X_i$, $S^2=\frac{1}{n-1}\sum(X_i-\bar{X})^2$, $X_{(1)}\le X_{(2)}\le\cdots\le X_{(n)}$, $M=\begin{cases}X_{[(n+1)/2]}, & n\text{-odd}\\ (X_{[n/2]}+X_{[n/2+1]})/2, & n\text{-even}\end{cases}$

(100)pth percentile: $\begin{cases}X_{(\{np\})}, & \frac{1}{2n}<p<0.5\\ X_{(n+1-\{n(1-p)\})}, & 0.5<p<1-\frac{1}{2n}\end{cases}$ · $S^2=\frac{1}{n-1}\sum(X_i-\mu)^2-\frac{n}{n-1}(\bar{X}-\mu)^2$

Thm. Let $X_1,\dots,X_n$ w/ $\mu,\sigma^2<\infty$. Then $E(\bar{X})=\mu$, $\mathrm{Var}(\bar{X})=\sigma^2/n$, $E(S^2)=\sigma^2$

Thm. Let $X_1,\dots,X_n\sim N(\mu,\sigma^2)$. Then $\bar{X}\sim N(\mu,\sigma^2/n)$, $\bar{X}\perp\!\!\!\perp S^2$, $\frac{(n-1)S^2}{\sigma^2}\sim\chi^2_{n-1}$

Thm. Suppose $X_{(1)}<\cdots<X_{(n)}$. $f_{X_{(1)},\dots,X_{(n)}}(y_1,\dots,y_n)=n!\,f_X(y_1)\cdots f_X(y_n)\,I\{y_1<\cdots<y_n\}$, $(y_1,\dots,y_n)=(X_{(1)},\dots,X_{(n)})$

$f_{X_{(j)}}(y)=\frac{n!}{(j-1)!(n-j)!}f_X(y)[F_X(y)]^{j-1}[1-F_X(y)]^{n-j}$; $f_{X_{(i)},X_{(j)}}(u,v)=\frac{n!}{(i-1)!(j-1-i)!(n-j)!}f_X(u)f_X(v)[F_X(u)]^{i-1}\times[F_X(v)-F_X(u)]^{j-1-i}[1-F_X(v)]^{n-j}I(u<v)$, $-\infty<u<v<\infty$

Prop. Let $R=X_{(n)}-X_{(1)}$: $f_R(r)=n(n-1)r^{n-2}(1-r)$

Prop. Let $U=X_{(1)}$, $V=X_{(n)}$; for $u<v$, $f(u,v)=n(n-1)[F(v)-F(u)]^{n-2}f(u)f(v)$

$X_n\xrightarrow{P}X := \lim_{n\to\infty}P(|X_n-X|<\varepsilon)=1=\lim_{n\to\infty}P(|X_n-X|\ge\varepsilon)=0\ \forall\varepsilon>0$

$X_n\xrightarrow{as}X := P(\lim_{n\to\infty}|X_n-X|<\varepsilon)=1\ \forall\varepsilon>0\equiv P(\lim_{n\to\infty}|X_n-X|=0)=1=P(\lim_{n\to\infty}X_n=X)=1$

Chebychev: X-RV, $g(x)\ge 0\ \forall x$: $P(g(X)\ge r)\le\frac{E(g(X))}{r}$; Let $g(x)=(x-E(x))^2$: $P(|X-E(X)|\ge t)\le\frac{\mathrm{var}X}{t^2}$; $g(x)=\frac{(x-\mu)^2}{\sigma^2}\Rightarrow P(|X-\mu|\ge t\sigma)\le\frac{\mathrm{var}X}{t^2\sigma^2}=\frac{1}{t^2}$

WLLN: $X_1,\dots,X_n$ iid w/ $\mu,\sigma^2<\infty$, then $\bar{X}_n\xrightarrow{P}\mu$; SLLN: $\bar{X}_n\xrightarrow{as}\mu$

Thm. $X_1,\dots,X_n$ iid w/ μ,σ^2, $\mu_4=E[(X-\mu)^4]<\infty$, then $S^2\xrightarrow{P}\sigma^2$; $E(S^2)=\sigma^2$, $\mathrm{var}(S^2)=\frac{1}{n}\left[\mu_4-\frac{n-3}{n-1}\sigma^4\right]$

$X_n\xrightarrow{d}X := \lim_{n\to\infty}F_n(x)=F(x)$ where F is continuous @ x

Prop. · If pdf/pmf/mgf of X_n converges to pdf/pmf/mgf of X, then $X_n\xrightarrow{d}X$

· $X_n\xrightarrow{P}X\Rightarrow X_n\xrightarrow{d}X$

· If $X_n\xrightarrow{d}X$ is degenerate (X has single possible value), then $X_n\xrightarrow{P}X$

Thm. $P(|X_n-\mu|>\varepsilon)\to 0\ \forall\varepsilon>0\equiv P(X_n<\varepsilon)\to\begin{cases}0, & x<\mu\\ 1, & x>\mu\end{cases}$ where μ: constant

CLT: Let $X_1,\dots,X_n\sim F$ w/ μ,σ^2. Then $\sqrt{n}(\bar{X}_n-\mu)\xrightarrow{d}N(0,\sigma^2)\equiv\frac{\sqrt{n}(\bar{X}_n-\mu)}{\sigma}\xrightarrow{d}N(0,1)$

$\mathrm{Bin}(n,p)\xrightarrow{d}N(np,np(1-p))$, Taylor series: infinite power series representation of smooth function $f(x)$ around a; uses derivs for a local polynomial approx: $\sum_{n=0}^{\infty}\frac{f^{(n)}(a)}{n!}(x-a)^n$

· $\left(1+\frac{a_n}{n}\right)^n\to e^{a_n}$ and $\sum_{n=0}^{\infty}a_n/n!\to e^{a_n}$ as $n\to\infty$

Thm (cont. mapping): a) $X_n\xrightarrow{d}X$ and h: cont. func. on support of dist. of X. Then $h(X_n)\xrightarrow{d}h(X)$

b) $X_n\xrightarrow{P}X$ and h: cont func. " " " ". Then $h(X_n)\xrightarrow{P}h(X)$

Thm (Slutsky's): $X_n\xrightarrow{d}X$, $Y_n\xrightarrow{d}a\in\mathbb{R}\Rightarrow Y_nX_n\xrightarrow{d}aX$ and $X_n+Y_n\xrightarrow{d}X+a$

Thm (Delta meth): If g is a diff. func. at μ with $g'(\mu)\ne 0$, then $\sqrt{n}[g(\bar{X}_n)-g(\mu)]\xrightarrow{d}N(0,[g'(\mu)]^2\sigma^2)$ (assume CLT holds) | Assume $g'(\mu)=0$, $g''(\mu)\ne 0$: $n[g(\bar{X}_n)-g(\mu)]\xrightarrow{d}\frac{1}{2}\sigma^2g''(\mu)\cdot\chi^2_1$

variance-stabilizing transformation: $[g'(\mu)]^2=$ constant | set to 1 unless told otherwise; limiting var does not depend on unknown μ.

Fisher's z-transformation: $(X,Y)\sim N(\cdot,\cdot)$ w/ ρ: corr. CLT shows $\sqrt{n}(r_n-\rho)\xrightarrow{d}N(0,(1-\rho^2)^2)$, via Delta, $\sqrt{n}[g(r_n)-g(\rho)]\xrightarrow{d}N(0,[g'(\rho)]^2(1-\rho^2)^2)$; Then $g(\rho)=\frac{1}{2}\log\left(\frac{1+\rho}{1-\rho}\right)\Rightarrow\sqrt{n}[g(r_n)-g(\rho)]\xrightarrow{d}N(0,1)$

Data Reduction: SS, Factorization Thm, Exponential Family → e.g. $N(\mu,\sigma^2)$, $\mathrm{Bin}(n,p)$

Stat model: $\mathcal{P}=\{P_\theta|\theta\in\Theta\}$ of possible prob dist. on $\mathcal{X}$ where P_θ: known, θ: unknown

· If $T(X)$ is a SS for θ, any inference abt θ should depend on X only through $T(X)$

Def: $T(X)$ is a SS if cond. dist. of $X|T(X)$ does not depend on θ

Thm (Factorization): $T(X)$ is a SS iff $\exists\ g_\theta(t)$ and $h(x)$ s.t. $f_\theta(x)=g_\theta[T(x)]h(x)$

Exp Family: $f(x|\theta)=h(x)c(\theta)\exp\left(\sum_{j=1}^{k}w_j(\theta)T_j(x)\right)$, $\theta=(\theta_1,\dots,\theta_d)$, $d\le k$

$\eta_j:=w_j(\theta)$; natural param of exp fam s.t. $f(x|\eta)=h(x)c(\eta)\exp\left(\sum_{j=1}^{k}\eta_jT_j(x)\right)$

Then $\prod_{i=1}^n f(x_i|\theta)=[c(\theta)]^n\prod h(x_i)\exp\left(\sum_{j=1}^{k}w_j(\theta)\sum_{i=1}^{n}T_j(x_i)\right)$

$\Rightarrow T(X)=\left(\sum_{i=1}^n T_1(X_i),\dots,\sum_{i=1}^n T_k(X_i)\right)$ is a SS for θ

Prop. If $\eta=\eta(\theta)$, $\theta\in\Theta\subset\mathbb{R}^d$, $d\le k$ s.t. $\{\eta(\theta):\theta\in\Theta\}$ is not open subset of $\mathbb{R}^k$, where $T(X)\in\mathbb{R}^k$, then this is curved EXP family

4.1: Minimal SS, Ancillary Statistics | 5.1 - Point Estimation: Method of Moments (MOM)

Def: $T(X)$ and $\tilde{T}(X)$ are equivalent if $\tilde{T}(X) = r\,T(X)$ for r: one-to-one transformation

Prop. $T(X)$ is sufficient i.f.f. $\tilde{T}(X)$ is sufficient

e.g. $T(X) = (\sum X_i, \sum X_i^2) = r(\bar{x}, s^2)$

Def: we say $T(X)$ is more informative if $\tilde{T}(X) = r\,T(X)$

Prop. If $T(X)$ is more informative than $\tilde{T}(X)$ and $\tilde{T}(X)$ is sufficient, then $T(X)$ is also sufficient

- For data reduction, simpler SS are more useful

Def: $T(X)$ is minimal SS (for θ) if $T(X)$ is sufficient and is a function of any other SS

Thm (Minimality): Given $f(x|\theta)$, if $\exists\, T(X)$ s.t. $\forall\, x, y$: $\frac{f(x|\theta)}{f(y|\theta)}$ doesn't depend on θ i.f.f. $T(x) = T(y)$ then $T(X)$ is a minimal SS for θ.

Ancillary stat: a stat $T(X)$ whose dist. does not depend on θ

- not relevant alone; can provide info when combined w/ other stats

Prop (LS-Fam).

- Given $F_m(x) = f_0(x-m)$, the difference btwn two obs in the loc-fam removes the loc. param m; $z = x - m \iff x = z + m$
- Given $f_\sigma(x) = \frac{1}{\sigma} f_0(\frac{x}{\sigma})$, the ratio of two obs in the scale-fam removes the scale param σ; $z = x/\sigma \iff x = \sigma z$
- Given $f_{m,\sigma}(x) = \frac{1}{\sigma}(\frac{x-m}{\sigma})$, standardizing removes both m and σ; $z = \frac{x-m}{\sigma} \iff x = \sigma z + m$

4.2: Complete Stats, Basu's Lemma

Def: Let $\{f(t|\theta) : \theta \in \Theta\}$ be a fam. dist. for $T(X)$. The family of prob. dist.'s is complete and $T(X)$ is a complete stat if $\forall$ function g:

$E_\theta[g(T(X))] = 0 \Rightarrow P_\theta[g(T(X)) = 0] = 1$
$\iff g(T) \xrightarrow{a.s.} 0$ for all θ

Def: A complete SS is both complete & sufficient

Prop. If T is ancillary, then dist. of T doesn't depend on θ; hence
$E[g(T)] = c \Rightarrow E_\theta[g(T) - c] = 0 \not\Rightarrow g(T) - c = 0$

Thm (Complete Stat in Exp Family): Let $X_i, i = 1, \dots, n$ iid, $f(x|\theta) = h(x)c(\theta)\exp(\sum_{i=1}^k w_i(\theta) T_i(x))$, $\theta \in \mathbb{R}^k$, then $T(X) = (\sum_{i=1}^n T_1(x_i), \dots, \sum_{i=1}^n T_k(x_i))$ is complete if parameter space Θ contains an open set in $\mathbb{R}^k$

Remarks: not an i.f.f. statement

- If # free params in $\theta < k$, then we can't identify an open set in $\mathbb{R}^k$

Thm (Basu's): If $T(X)$ is complete and SS for θ, then $T(X) \perp\!\!\!\perp$ every ancillary stat $S(X)$

Thm: If $T = T(X)$ is complete and sufficient for θ, then T is minimal sufficient for θ.

Corollary: If a minimal SS $\exists$, then any complete stat is not necessarily also a minimal SS

$\sigma^2 = E(x^2) - E^2(x)$ FTC: $\frac{d}{dx}\int_a^\infty f(t)\,dt = f(x)$

$a^x a^y = a^{x+y}$, $\frac{a^x}{a^y} = a^{x-y}$, $(a^x)^y = a^{xy}$, $(ab)^x = a^x b^x$

Point estimation: estimating unknown parameter by a number/vector

Estimator (procedure) $\hat\theta(X)$ + sample $\rightarrow$ Estimate $\hat\theta(x)$; function of data; RV w/ a sampling dist.

- In general, compute discrepancy measures based on sampling dist. (no simulation required)

MOM: Assume random sample $X_1, \dots, X_n$ from a dist. that is indexed by $\theta \in \mathbb{R}^k$ w/ at least k-finite moments

- obtain $\hat\theta_{MOM}$ by equating first k sample moments to corresponding k theoretical (pop) moments, and solving the resulting simultaneous equation:

$m_j = \frac{1}{n}\sum_{i=1}^n X_i^j$ and $m_j'(\theta) = E_\theta(X^j)$, $j = 1, \dots, k$; solve system of equations via substitution or elimination: $m_j'(\theta) = m_j$, $j = 1, \dots, k$

Generalized MOM: Define $g(\theta) = E_\theta(X)$, g: 1-1.
Then $\frac{1}{n}\sum X_i = g(\theta) \iff \frac{1}{n}\sum(X_i - g(\theta)) = 0$, let $\psi_\theta(x) = x - g(\theta)$
s.t. $\frac{1}{n}\sum \psi_\theta(x_i) = 0 = E_\theta[\psi_\theta(X)]$.

Generalize via 1) picking $\psi_\theta(x)$ that depends on θ and satisfies $E_\theta[\psi_\theta(X)] = 0$ for all θ
2) estimating θ by solving $\frac{1}{n}\sum \psi_{\hat\theta}(x_i) = 0$

5.2: Maximum Likelihood Estimation (MLE); Point Est.

$L(\theta|x)$: x obs, θ varying (over all possible params)
$=$
$f(x|\theta)$: θ fixed (unknown), x variable

- If $P_{\theta_1}(X = x) = L(\theta_1|x) > L(\theta_2|x) = P_{\theta_2}(X = x)$, then sample more likely under θ_1 than θ_2

$\hat\theta_{MLE}(X) = \underset{\theta \in \Theta}{\operatorname{argmax}}\, L(\theta|x)$ (estimator) (*) and $\hat\theta_{MLE}(x)$: estimate

- intuitive; range of param. coincides w/ MLE range
- Focus on closed form (analytical) solution for (*)

Approach: maximize $\ell(\theta|x)$; a strictly increasing function on $(0, \infty)$ and hence extrema of $L(\theta|x) \equiv$ extrema of $\ell(\theta|x)$.

$L(\theta|x) = \prod f(x_i|\theta)$; $\ell(\theta|x) = \sum \log f(x_i|\theta)$; $\ell'(\theta) = \sum \frac{d}{d\theta}\log f(x_i|\theta) \overset{set}{=} 0 \rightarrow$ solve for $\hat\theta_{MLE}$
ensure max achieved @ saddle point via $[d\ell^2/d\theta^2] < 0$

- Local max of: $g_i(x) = \frac{\partial}{\partial x_i} g(x)$, $g_{ij}(x) = \frac{\partial^2}{\partial x_i \partial x_j} g(x)$; solve $\nabla g(x) = (g_1(x), \dots, g_m(x))^T \overset{set}{=} 0$ for $\hat{x} \in \mathbb{R}^m$ ($g(x_1, \dots, x_m)$)
verify $H = \{g_{ij}(x)\}_{i,j=1}^m$ is strictly neg def: $-H^T H \hat{x} < 0$ $\forall\, \hat{x} \neq 0$ (all eigs are strictly negative)

Exp Fam: Provided full rank, i.e. $f(x|\theta) = h(x)c(\theta)\exp(\sum_{i=1}^k w_i(\theta) t_i(x))$ (no linear constraint)
then $\ell(\theta|x) = \log h(x) + \log c(\theta) + \sum w_i(\theta) t_i(x)$; $\partial \ell(\theta) = \frac{c'(\theta)}{c(\theta)} + \sum w_i'(\theta) t_i(x) \overset{set}{=} 0$ $\rightarrow$ solve for $\hat\theta_{MLE}$

Limitations: (1) $\hat\theta_{MLE} \notin$ open set of param space Θ, (2) Non-uniqueness of $\hat\theta_{MLE}$

Invariance: Provided $\hat\theta$: MLE of θ, $g(\hat\theta)$: MLE of $g(\theta)$ for some arbitrary function $g(\theta)$

- Sometimes no closed form solutions; Newton's Method: solve $f(\theta) = 0$, let θ_0: initial guess, then $\theta_{n+1} = \theta_n - \frac{f(\theta_n)}{f'(\theta_n)}$, $n \geq 0$, updating solution $\hat\theta$ until x_n converges

5.3: Bayes Estimators

- MLE: $f(x|\theta)$, θ: unknown/fixed
- Bayesian: treat θ: unknown/random

$\pi(\theta)$: prior; formulated before the data has been seen
$f(x, \theta) = f(x|\theta)\pi(\theta)$: joint dist. of (x, θ)
$m(x) = \int_\Theta f(x, \theta)\,d\theta$: (marginal dist. of X) from unknown data is generated

- Prior is updated w/ sample (x) from $m(x)$ and called the posterior dist

$\pi(\theta|x) = \frac{f(x, \theta)}{m(x)} = \frac{f(x|\theta)\pi(\theta)}{m(x)} \propto f(x|\theta)\pi(\theta) = L(\theta|x)\pi(\theta)$ — dependence on θ; helps to identify form (dist.) of $\pi(\theta|x)$

- $L(\theta|x) = f(x|\theta)$ - just interpreted differently. drop anything w/o

kernel: part of density that depends on parameter θ; ignoring multiplicative constants
$\rightarrow$ helps to simplify $\pi(\theta|x)$ calculation

conjugate Priors: specific choice of $\pi(\theta)$ that guarantees same form (dist) for $\pi(\theta|x)$

e.g.s.
- $X_i \sim Bern(\theta)$, $\theta \sim Beta(\alpha_0, \beta_0)$, $\theta|x \sim Beta(\alpha_0 + \sum x_i, \beta_0 + n + \sum x_i)$
- $X_i \sim Pois(\theta)$, $\theta \sim Gam(\alpha_0, \beta_0)$, $\theta|x \sim Gam(\alpha_0 + \sum x_i, \beta_0 + n)$
- $X_i \sim N(\theta, \sigma^2)$, $\theta \sim N(m_0, v_0^2)$, $\theta|x \sim N\left(\frac{n\bar{x}_n v_0^2 + \sigma^2 m_0}{n v_0^2 + \sigma^2}, \frac{\sigma^2 v_0^2}{n v_0^2 + \sigma^2}\right)$

Approach: systematically invent conjugate priors for $f(x|\theta)$ that are exp fam's:

1) $f(x_i|\theta) = h(x_i)c(\theta)\exp[w(\theta)T(x_i) - A(\theta)]$; $f(x|\theta) \propto \exp[w(\theta)\sum T(x_i) - nA(\theta)]$ (*)
2) kernel of prior $\pi(\theta) \propto \exp[w(\theta)\chi_0 - \nu_0 A(\theta)]$ (F); introduce new hyperparams χ_0, ν_0
3) $\pi(\theta|x) \propto f(x|\theta)\pi(\theta) = \exp[w(\theta)(\chi_0 + \sum T(x_i)) - (\nu_0 + n)A(\theta)]$ (same form as $\pi(\theta)$ w/ new params)

5.3 (Bayes Est. cont.)

Est: Extract $\hat{\theta}$ from $\pi(\theta|x)$ via $L(\theta, a)$: quantifies difference btwn θ and param est. a

Bayes Risk: $E[L(\theta,a)|x] = \int_{\Theta} L(\theta,a)\pi(\theta|x)\,d\theta$

where $\hat{\theta}_{Bay} = \arg\min_a E[L(\theta,a)|x]$

(square error)
$L(\theta,a) = (\theta-a)^2 \rightarrow \hat{\theta}_{Bay} = E(\theta|x)$

P: $\frac{\partial}{\partial a} E[(\theta-a)^2|x] = -2E(\theta|x) + 2a \overset{set}{=} 0 \Rightarrow \hat{\theta} = E(\theta|x)$. ▫

(abs error)
$L(\theta,a) = |\theta - a| \rightarrow \hat{\theta}_{Bay}$ = med. of $\pi(\theta|x)$

$L(\theta,a) = I(\theta \neq a) \rightarrow \hat{\theta}_{Bay}$ = mode of $\pi(\theta|x)$

Prop. $\hat{\theta}_{Bay}$ based on $(\theta-a)^2$ for $g(\theta)$ is posterior mean of $g(\theta)$

- $\hat{\theta}_{Bay}$ not transformation invariant
- $\pi(\theta)$ plays little role for large n

5.4: Evaluating Estimators

$Bias(\hat{\theta}) := E(\hat{\theta}) - \theta = 0 \Rightarrow$ unbiasedness

$var(\hat{\theta})$: $E[(\hat{\theta} - E(\hat{\theta}))^2]$

$MSE(\hat{\theta}) := E[(\hat{\theta}-\theta)^2] = \{Bias(\hat{\theta})\}^2 + var(\hat{\theta})$

P. $E[(\hat{\theta}-\theta)^2] = E[(\hat{\theta}-E(\hat{\theta})) + (E(\hat{\theta})-\theta)]^2 =$
$[E(\hat{\theta})-\theta]^2 + E[(\hat{\theta}-E(\hat{\theta}))]^2$
$+ 2E[(\hat{\theta}-E(\hat{\theta}))][E(\hat{\theta})-\theta]$
$2[E(\hat{\theta})-\theta][E(\hat{\theta}) - \cancel{E(\hat{\theta})}]$. ▫

Suppose $\theta = (\theta_1, \dots, \theta_k)^T$. Then

$Bias(\hat{\theta}) = (E(\hat{\theta}) - \theta) \in \mathbb{R}^k$

$var(\hat{\theta}) = E[(\hat{\theta}-E(\hat{\theta}))(\hat{\theta}-E(\hat{\theta}))^T]$
$= cov(\hat{\theta}) \in \mathbb{R}^{k \times k}$

$MSE(\hat{\theta}) = Bias(\hat{\theta})Bias(\hat{\theta})^T + var(\hat{\theta})$

5.5: Best unbiased Estimators

RB (Thm): $T(X)$: SS for θ, $E(\hat{\theta}) = \theta$. Then $\hat{\theta}_{RB} := E(\hat{\theta}|T(X))$ and $E(\hat{\theta}_{RB}) = \theta$, $var(\hat{\theta}_{RB}) \le var(\hat{\theta})\ \forall \theta$: preserve unb., filter out noise

BUE: $\hat{\theta}$ is BUE if $var(\hat{\theta}) \le var(\tilde{\theta})$ for all θ and unbiased est's $\tilde{\theta}$

LS (Thm). Let $T(X)$: complete SS. Then $E[\phi(T)] = \theta \Rightarrow \phi(T)$ is $\hat{\theta}_{BUE}$ (unique).

- Extend RB and LS for $b(\theta)$ for some function b.

Fisher info - $\theta \in \mathbb{R}^1$ is scalar

$S(\theta) = \frac{\partial}{\partial\theta}\log f(x|\theta)$; $E[S(\theta)] = 0$

$I_x(\theta) = E[\{S(\theta)\}^2] = -E\left\{\frac{\partial^2}{\partial\theta^2}\log f(x|\theta)\right\}$ (*)

$nI_x(\theta) = I_X(\theta)$ for x: obs, X: sample

(*) under reg cond: $\int \frac{\partial^2}{\partial\theta^2} f(x|\theta)\,dx = 0$

CRI: Let $E(T) = b(\theta)$. Then
$var[T(X)] \ge \frac{[b'(\theta)]^2}{nI_x(\theta)}$

- efficient est achieves CRLB

Recall (BSI). $[cov(U,V)]^2 \le var(U)var(V)$
$\Leftrightarrow var(U) \ge [cov(U,V)]^2/var(V)$

Prop. Est achieves CRLB $\Leftrightarrow$ linear in $S(\theta)$

Fisher info - $\theta \in \mathbb{R}^k$

$S(\theta) = \partial \log f(x|\theta) = \begin{bmatrix} \frac{\partial}{\partial\theta_1}\log f(x|\theta) \\ \vdots \\ \frac{\partial}{\partial\theta_k}\log f(x|\theta) \end{bmatrix} \in \mathbb{R}^k$

$E[S(\theta)] = 0 \in \mathbb{R}^k$

$I(\theta) = E[S(\theta)S(\theta)^T|\theta) \in \mathbb{R}^{k\times k}$

where $I(\theta)_{i,j} = E[(\frac{\partial}{\partial\theta_i}\log f(x|\theta))(\frac{\partial}{\partial\theta_j}\log f(x|\theta))]$

+ diag elements: 1D Fisher info $\forall\ \theta_1, \dots, \theta_k$

$I(\theta)_{i,j} = -E\left\{\frac{\partial^2}{\partial\theta_i\partial\theta_j}\log f(x|\theta)\right\} : -E\begin{bmatrix} \frac{\partial^2}{\partial\theta_1^2}\log f(x|\theta) & \cdots & \frac{\partial^2}{\partial\theta_1\partial\theta_k}\log f(x|\theta) \\ \vdots & & \\ \frac{\partial^2}{\partial\theta_k\partial\theta_1}\log f(x|\theta) & \cdots & \frac{\partial^2}{\partial\theta_k^2}\log f(x|\theta) \end{bmatrix}$

$I_x(\theta) \cdot n = I(\theta)$

$\theta = (\theta_1, \dots, \theta_k)^T$

CRLB: Let $E[T(X)] = g(\theta) \in \mathbb{R}^m$, $J(\theta) = \left[\frac{\partial g_i(\theta)}{\partial\theta_j}\right]_{m\times k}$. Then

$cov[T(X)] \succeq J(\theta)I(\theta)^{-1}$ for all θ | $A \succeq B \Rightarrow A-B$: eigs ≥ 0, i.e. $A-B$: positive semi-definite

a) Let $g(\theta), T(X)$: scalar. Then $cov[T(X)] \ge [\nabla g(\theta)]^T I(\theta)^{-1}[\nabla g(\theta)]$

b) Let $g(\theta) = b(\theta_1)$ and $(\theta_2, \dots, \theta_k)$: nuisance params

+ $(\theta_2, \dots, \theta_k)$ known $\Rightarrow var[T(X)] \ge [b'(\theta_1)]^2/nI_x(\theta)$
+ $(\theta_2, \dots, \theta_k)$ unknown $\Rightarrow cov[T(X)] \ge [b'(\theta_1), 0, \dots, 0]^T I(\theta)^{-1}[b'(\theta_1), 0, \dots, 0]$
$= [b'(\theta)]^2 I(\theta)^{-1}_{11}$

where $I_{11}(\theta)^{-1} = \frac{1}{I_{11\cdot2}(\theta)} = \frac{1}{I_{11} - I_{12}I_{22}^{-1}I_{21}} = I_{11}(\theta)$ if $I_{12}(\theta) = 0$

$\le I_{11}(\theta) \Rightarrow [b'(\theta_1)]^2/I_{11}(\theta) \le [b'(\theta_1)]^2/I_{11\cdot2}(\theta)$

i.e. final var LB is larger from uncertainty via nuisance params

MLE (asymptotic property): obtain $\hat{\theta}_{MLE}$ via $S(\theta) = \frac{\partial \log f(x|\theta)}{\partial\theta} \overset{set}{=} 0$

Then $\sqrt{n}(\hat{\theta}_{MLE} - \theta) \xrightarrow{d} N(0, [I(\theta)]^{-1}) = N(0, \frac{1}{I(\theta)}) = I^{-1}(\theta)N(0, I(\theta))$

- MLE asy. efficient; achieves CRLB as $n \to \infty$